

Cash or card? A structural model of payment choices^{*}

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January 21, 2026

Abstract

We use a large granular dataset to analyze the households' choice between cash and card payments. Empirically, both the *size of the transaction* and the amount of *cash on hand* appear as significant covariates of the payment choice. We unveil a novel interaction between these two variables: the critical size for a card purchase depends on the amount of cash on hand. We present a tractable model of payment choices, featuring non-universal acceptance of cards by merchants, and a random expenditure flow. The model generates a precautionary motive for holding a cash buffer: cards are used to avoid “running out of cash”, accounting for the interaction discussed before. We use a calibrated version of the model to quantify the benefits of card ownership, the welfare costs of imperfect card acceptance by merchants, and to identify conditions under which a *cashless economy* emerges.

Keywords: *cash management, payment choices, inventory model, card payments.*

JEL classifications numbers: *E41, E42, D14.*

^{*}We thank Fernando Alvarez, Russell Cooper, Martin Brown, Alessandro Secchi, Elena Falcettoni and Alejandro Zamora-Pérez for useful comments. We benefited from the feedback of seminar and conference participants at EIEF, Luiss, the Bank of Italy, the European Central Bank, the European University Institute, Economics of Payments XIV and the 2025 CEBRA Annual Meeting. Lippi acknowledges financial support from the ERC grant: 101054421-DCS. The views expressed in this paper are those of the authors and do not necessarily reflect the views of the Bank of Italy. E-mail: francescolippi@gmail.com, elia.moracci@bancaditalia.it.

1 Introduction

Advances in information technologies and digitization change the payment behavior of households and raise interesting questions on the welfare benefits of new technologies, and the future of cash (see [Trütsch \(2020\)](#), [Shy \(2023\)](#), and [Nocciola and Zamora-Peréz \(2024\)](#)). We present new evidence on means-of-payment choices in the euro area, drawn from surveys conducted by the European Central Bank between 2015 and 2024. The transaction-level data shows that the households' *choice* on whether to pay with cash vs card, in situations where both options are feasible, deviates from the predictions of simple means-of-payment models. First, for a substantial fraction of individuals the largest reported cash payment is bigger than the smallest card transaction occurred on the same day, a behavior that contradicts the simple *threshold-rule* prediction by [Whitesell \(1989\)](#). Second, card payments are often chosen when households have sufficient cash to carry out the transaction, contradicting *cash burns* policies that prescribe using cards only after running out of cash, as in [Alvarez and Lippi \(2017\)](#). Moreover, the data reveals a novel pattern in payment behavior: card usage becomes more likely as the size of the purchase approaches the level of cash on hand, i.e., when the residual money holdings in case of a cash payment are low. The latter is consistent with a precautionary motive: a cash buffer provides insurance for the transactions where cards cannot be used.

We propose a dynamic model of cash management to interpret the facts. The model features a stochastic expenditure flow and imperfect acceptance of cards by merchants. The agent chooses the frequency and size of cash withdrawals, and can settle her purchases with cash or with a card. As customary in inventory-theoretic models, cash withdrawals entail a fixed cost; we also assume that using the card entails a fixed cost per-purchase.¹ The model solution prescribes the optimal payment policy: the agent holds a precautionary stock of cash to avoid paying excessive card fees, as well as to insure against shopping trips in which cards are not accepted. If the cost of using cards relative to cash is small, the agent uses the card even when cash on hand is sufficient, in order to preserve her cash buffer. The optimal policy shares features of [Whitesell \(1989\)](#) – namely, for a given level of cash balances, the

¹This cost represents fees and the non-pecuniary costs of using cards, among which privacy concerns.

agent follows a transaction-size threshold policy – and of [Alvarez and Lippi \(2017\)](#) – namely, that cash is always employed to settle small-value transactions. The optimal policy features a novel interaction between the amount of cash on hand and the size of the purchase: the transaction-size threshold above which the agent uses cards depends on the amount of cash held before the payment occurs.

Our analysis rests on three important modeling ingredients. First, the random timing and sizes of expenditures create the possibility that the agent may find herself without enough cash to carry out a purchase, giving rise to a precautionary demand for cash, as in [Telyukova \(2013\)](#).² Second, the choice between cash versus card payments is modelled sequentially, so that the agent chooses whether to use cash or card at each moment, conditional on the purchase size and her cash holdings. This is similar to [Alvarez and Lippi \(2017\)](#) and [Briglevics and Schuh \(2021\)](#), and differs from the classic analysis by [Lucas and Stokey \(1987\)](#) and [Whitesell \(1989\)](#) where the agent simultaneously chooses the mix between cash versus card payments over a fixed time period.³ Third, the setup features (exogenous) imperfect acceptance of cards by merchants. This assumption boosts the precautionary motive for holding a cash buffer, a feature that is strongly apparent in the data.

We calibrate the model by matching selected moments on cash management and payment choices for the euro area households, computed from diaries collected by the ECB in 2023-2024. We use the calibrated model to inspect features of the current payment system and develop some counterfactual analyses. The calibrated model reveals an overall annual cost of managing transactions that is remarkably small (around 15 euros for the average household), suggesting that the payment system imposes a relatively small burden on the households' budgets. We analyze a counterfactual scenario where households can only use cash, and find that the annual cost of managing transactions is about 50 euros, i.e. that the value of a payment card is around 35 euros per year, highlighting the quantitative importance

²The assumption contrasts with the standard inventory models where consumption follows a continuous path, see [Miller and Orr \(1966\)](#), [Eppen and Fama \(1969\)](#), and [Alvarez and Lippi \(2009\)](#), among others. Technically, consumption follows a compound Poisson process, see [Perera and Sethi \(2023\)](#) for a review. Examples of cash management models including (or exclusively making use of) compound Poisson expenditure include [Beckmann \(1961\)](#) and [Bar-Ilan, Perry, and Stadjé \(2004\)](#).

³In those classic models one cash withdrawal occurs in every period, and all cash is spent. No purchase opportunities are ever missed and statistics such as the size and frequency of withdrawals are exogenously determined by the model's time period length.

of cards for consumers' welfare. We also show that a complete ban of cash would increase the cost of managing consumption transactions by around 60 euros per year for the average consumer.⁴ We estimate that the imperfect acceptance of cards by merchants is costly for households: under near-universal acceptance of cards, the annual cost of managing cash and settling transactions would decrease by almost 75% (from 15 to 4 euros per year). We use the model to identify the conditions for a cashless economy to emerge. The analysis shows that cash is quite resilient: increasing the rate of card acceptance from the current level of 89% to *near-universal* (99%) does not eliminate cash usage; the share of cash expenditures (over total consumption) falls substantially but remains non-negligible. Under near-universal acceptance, a cashless economy only emerges if card payments become as cheap as cash ones. Given the current card acceptance rate, even cards becoming substantially cheaper than cash would *not* lead to a cashless economy, as consumers would still hold a cash buffer to avoid missing a purchase.

1.1 Related literature

Several studies use micro-level evidence such as payment diaries or stores' transaction data to study the determinants of consumers' payment method choices (see [Shy \(2023\)](#) for a review of the literature). A first set of papers establishes a link between the payment method decisions and the size of the purchase. Exploiting grocery store data, [Klee \(2008\)](#) finds that cash is mainly used for small-sized purchases, while card payments are prevalent when the value of the sale increases. Similar results are obtained by [Wang and Wolman \(2016\)](#) leveraging on scanner data from two billion retail transactions, and by many other studies. A second relevant determinant of cash/card choices is the amount of cash on hand at the moment when the transaction is settled. The papers that analyzed the relationship between cash holdings and payment choices consistently found that the likelihood of cash usage increases with the level of cash holdings.⁵ A third potential determinant of payment choices by individuals are also supply-side factors such as limited acceptance of means of payments:

⁴A ban on cash is endorsed by [Rogoff \(2017\)](#); see [Alvarez, Argente, Jimenez, and Lippi \(2022\)](#) for a critical assessment of this proposal using Mexican data on crime and cash usage.

⁵See [Arango, Huynh, Fung, et al. \(2012\)](#), [Bouhdaoui and Bounie \(2012\)](#), [Huynh, Schmidt-Dengler, and Stix \(2014\)](#) and [Bagnall et al. \(2016\)](#).

several studies (see [Arango, Huynh, and Sabetti \(2015\)](#) and [Huynh, Schmidt-Dengler, and Stix \(2014\)](#), among others) showed that the probability of card acceptance by merchants affects means of payments decisions by consumers. Moreover, a number of studies (see for example [Bagnall et al. \(2016\)](#)) documented a relationship between payment choices and cash management decisions, finding a positive association between cash usage, the frequency and size of withdrawals, and average cash holdings. We contribute to this empirical literature using a rich payment diary dataset for the euro area over the years 2015 - 2022 and highlight a novel finding: households' choices to pay with cash or card are affected by how large the incoming transaction is *relative* to the amount of cash holdings.

A few theoretical studies analyze the households' payments choice using the standard cash management model à la [Baumol \(1952\)](#) and [Tobin \(1956\)](#) (BT from now on). [Whitesell \(1989\)](#) includes a non-degenerate distribution of transaction sizes in the BT model, as well as a choice among cash and card payments. In this model, it is optimal to pay by cash whenever the size of the transaction is smaller than a given threshold, and to pay using cards otherwise (*transaction size threshold* policy). [Alvarez and Lippi \(2017\)](#) abstract from transaction size heterogeneity and present a dynamic model where payment choices only depend on the level of cash holdings, showing that cards are used only when the agent does not have enough cash at hand (*cash burns* policy). Notice that the predictions of both these models are broadly consistent with the above-cited empirical findings on the relationship of payment choices with transaction sizes and cash holdings. A related analysis by [Briglevics and Schuh \(2021\)](#) considers a model in which the agent's decision on how to pay depends both on the size of the transaction and on the amount of cash held, within a dynamic cash management framework, and estimated with US payment diary data.

1.2 Structure of the paper

The paper is organized as follows. [Section 2](#) presents our empirical analysis and highlights new stylized facts on the interplay between cash management and payment choices. [Section 3](#) outlines our theoretical framework and discusses the properties of the policy followed by the agent. We show how the model's solution maps into observable cash management and payment choice statistics. In [Section 4](#) we discuss our calibration strategy, discuss the results,

evaluate the model fit and perform a welfare analysis. We then move to applications. In [Section 5.1](#) we use our calibrated model to evaluate the welfare benefit of having a payment card. In [Section 5.2](#) we quantify the cost for households stemming from imperfect acceptance of cards, and study a counterfactual scenario with near-universal acceptance by merchants. In [Section 5.3](#) we assess the resilience of physical cash as a means of payment. [Section 6](#) concludes.

2 Stylized facts about households' payment behavior

This Section presents the data used in the paper and the main empirical patterns that motivate the analysis. In [Section 2.2](#) we compare the predictions of existing models of payment choice with evidence drawn from transaction-level data. In [Section 2.3](#) we illustrate new facts on payment choices by households and on the relationship between card acceptance rates and cash management behavior.

2.1 Data sources

We use payment diaries and questionnaires from two ECB surveys: i) the *Survey on the use of cash by households in the euro area* (SUCH from now on), that contains data from 2015 and 2016, ii) the first three waves of the *Study on the payment attitudes of consumers in the euro area* (SPACE I, II and III from now on), carried out in 2019, 2021-22 and 2023-24, respectively.⁶ Individuals participating in the survey are asked to record in a diary their payments on a given day.⁷ For each transaction the diary records the type of store, the amount paid and the payment method employed (cash or a cashless option, that we broadly label as *card payments*).⁸ For each purchase, respondents are also asked if alternative payment methods were accepted at the point of sale. The level of cash holdings at the start of the diary day, as well as any

⁶Although the surveys were conducted in different years and changed name over time, they utilize a nearly identical diary-based methodology. This ensures that the variables we rely on are comparable across the entire 2015–2024 sample period.

⁷Payments include both point-of-sale (POS) transactions, i.e., normal purchases in stores, and payments made to other individuals (P2P), such as charity or transfers to other family members, friends and colleagues. We disregard online payments as they are almost always settled using cards.

⁸We refer to all payment methods excluding cash as *cards*. These include debit cards, credit cards (both with and without a contactless option), mobile payments, credit transfers, bank cheques and other payment types. [Figure A.1](#) displays how often each of these payment instruments is used within every wave.

TABLE 1: Cash management and payment choices by households: summary statistics.

	Time period			
	2015-16	2019	2021-22	2023-24
<i>Expenditure flow</i>				
Daily expenditure (EUR)	34	46	64	61
Size of payment (EUR)	19	28	33	32
N. daily transactions	1.8	1.7	1.9	1.9
<i>Households' cash management</i>				
Cash holdings (EUR)	60	83	85	82
Withdrawal size (EUR)	70	90	108	96
Pr(Withdraws cash)	0.11	0.13	0.22	0.26
<i>Sellers' acceptance of payment methods</i>				
Card accepted	0.72	0.80	0.85	0.89
<i>Features of transactions</i>				
Forced card	0.09	0.14	0.19	0.16
Forced cash	0.28	0.20	0.15	0.11
Unforced (both options possible)	0.63	0.66	0.66	0.73
<i>Households' payment method choices</i>				
Pr(Card)	0.25	0.32	0.44	0.49
Pr(Card Unforced)	0.26	0.27	0.37	0.45
Number of respondents	64,214	40,182	39,459	40,710
Number of transactions	117,487	67,765	76,320	77,201

Note: All rows display sample averages. The daily expenditure and number of daily transactions are lower bounds for their true values as respondents can report at most eight purchases on the payment diary. The amount of cash holdings is that reported at the start of the diary day. The probability of a cash withdrawal is relative to the day of the survey. Online payments are excluded. Only respondents with access to cards are considered. The final sample includes a total of 338,773 transactions (with cash or cards) performed by 184,565 individuals. Data from ECB payment diaries: SUCH (2016) and SPACE (2019, 2021-22 and 2023-24).

other cash inflow that occurs during the day are reported as well: in particular, respondents report whether they withdrew cash (from ATMs, via cashback, or simply by moving coins and banknotes held at home to their wallets) or received it from others (as cash income, or by family and friends). This information enables us to back up the level of cash holdings before each transaction takes place, for a large set of participants.⁹ Finally, respondents answer a survey questionnaire where they provide additional information on their cash management and payment habits, as well as on a large set of demographic characteristics.

The data enables us to take a close look at European households' decisions concerning payment method's adoption and usage (in particular, whether to hold cards or not, and when

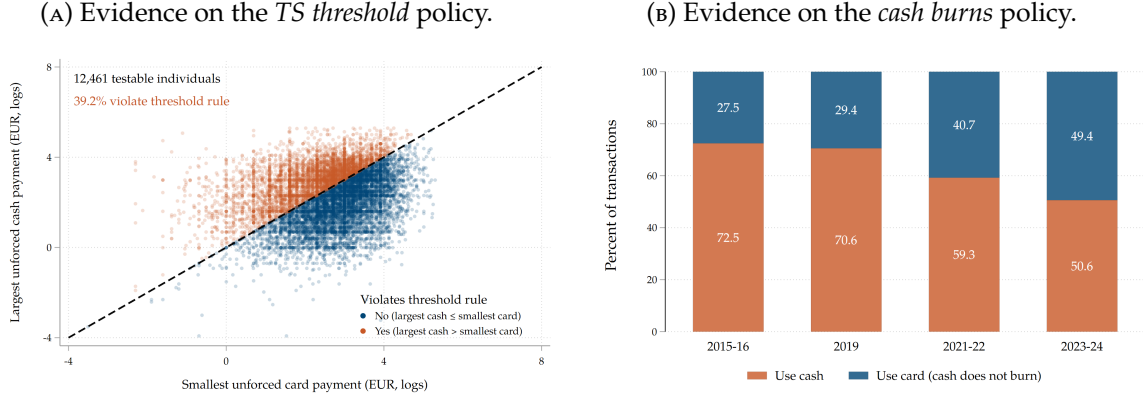
⁹See [Appendix A.2](#) for the technical details on the data cleaning process and on the derivation of additional information from the original datasets.

to use them), as well as on cash management policies (frequency and size of cash inflows and outflows) across the 2015-2024 period. A key feature is that having combined information on the level of cash holdings and the set of payment methods accepted in the point of sale, we can pin down the set of available payment options (*payment choice set*) for each transaction. This enables us to distinguish between payment choices that are *forced*, e.g., paying using the card because cash on hand is insufficient, from those that are *unforced*. Unforced transactions are those in which both cash and card payments are possible, i.e., in which the amount of cash holdings is larger than the purchase size, the merchant accepts card and cash payments, and the consumer has access to a payment card. It is evident that unforced transactions are important to identify the determinants of individual payment choices.¹⁰ In [Table 1](#), we display selected summary statistics across the four waves for the subsample of respondents with access to cards (around 99% of consumers). The Table describes the expenditure flow of households during the diary day and cash management behavior by households: we report the average cash holdings, the proportion of respondents that withdrew cash during the day, as well as the average withdrawal size.¹¹ The middle panel of the Table depicts the acceptance of payment methods by sellers. Card acceptance by merchants increased during this period. In the bottom of the Table we display the fraction of transactions in which cash and cards were available options, as well as the share of unforced transactions. This combines the supply-side acceptance decision by merchants with the buyers' choices. The share of *unforced* transactions has slightly increased over the sample period from around 60% to more than 70% of total transactions. The total fraction of card payments went up throughout the sample period, and so did the share of unforced transactions settled using cards (from 25% to about 45%).

¹⁰Most studies in the literature instead either disregard the fact that one can pay by cash only when the size of the transaction is smaller or equal than current cash holdings, or operate under the assumption of full acceptance of all payment methods, or both.

¹¹The notable increase in daily expenditures and cash holdings over time is largely attributable to changes in the composition of the sample and in the timing of data collection within the year. We estimate our structural model using the data from the 2023-24 survey.

FIGURE 1: Patterns of payments behavior.



Note: An *unforced payment* is one where both cash and card were viable payment options. In Panel (a), each dot corresponds to one respondent. In total, we have 12,461 individuals who report both an unforced cash payment and an unforced card payment during the diary day. The dashed line is the 45 degree line. Panel (b) displays the distribution of payment choices for *unforced* transactions. We say that *cash does not burn* when a card payment is preferred to cash.

Data from ECB payment diaries: SUCH (2016) and SPACE (2019, 2021-22 and 2023-24).

2.2 Patterns of payment choice: data vs models

Two main mechanisms have been proposed to explain how households choose between cash and cards. A first hypothesis is that individuals choose how to pay depending on the size of the transaction. A second one is that consumers use either cash or cards depending on the amount of cash they have on hand. We compare the implications of these theories with observed behavior.

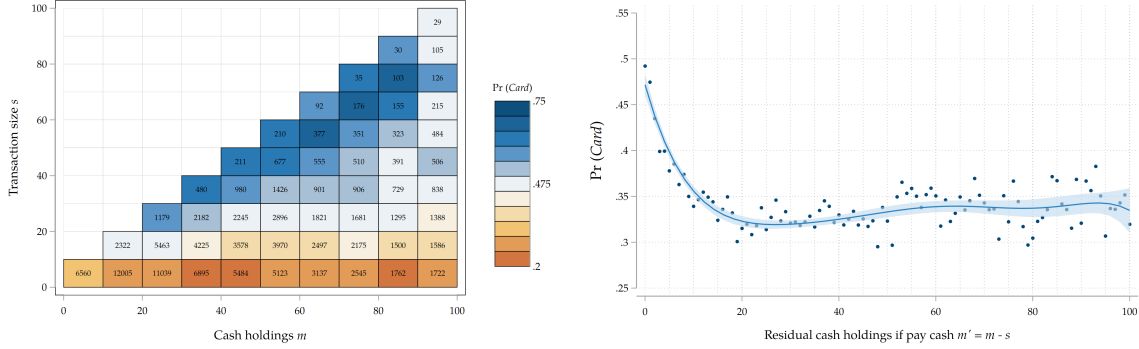
Transaction-size threshold policy. Some models emphasize the role played by the size of the transaction (which we denote by s) as a determinant of consumers' payment choices (Prescott (1987), Whitesell (1989), Freeman and Kydland (2000)).¹² Under the assumption that cash payments incur costs that are proportional to the size of the transaction, while cards required a fixed per-transaction cost, these papers derived optimal payment policies characterized by *transaction-size thresholds*: individuals settle using cash the expenditures which are smaller than some threshold transaction size ($s \leq \underline{s}$), while they exploit cards for large-sized purchases ($s > \underline{s}$). Support for this theoretical prediction has been provided by many empirical papers that find a negative correlation between the size of transactions faced

¹²Wang and Wolman (2016) offer a summary of this literature, which they refer to as the *threshold framework*.

and the probability of cash usage (Klee (2008), Wang and Wolman (2016)). However, the presence of a negative association is a rather qualitative test of the theory. Our data enables us to see if households actually adopt such threshold policies when deciding on how to pay: given that we observe multiple transactions for each individual, and that we can pin down the payment alternatives available at each transaction, we can compare the largest unforced cash payment and the smallest unforced card payment of respondents. Figure 1a reveals that support for threshold-based policies is quite weak: among the subset of households that reported both an unforced card payment and an unforced cash payment, around 39% of individuals explicitly violate the simple transaction-size threshold rule. This is contrast with the model’s predictions: even allowing for individual-specific thresholds s_i , under a transaction-size threshold rule all dots should lie below the 45 degree line.

“Cash burns” policy. Other models of payment choices focus on the level of cash holdings at the moment in which the transaction takes place (which we denote by m). Motivated by empirical findings showing that consumers are more likely to use cash as a means of payment when cash balances are larger,¹³ Alvarez and Lippi (2017) outline an inventory model augmented with a means of payment choice. In their setup, it is optimal to use cards only when $m = 0$, i.e., when the agent has no cash on hand at all: whenever she has cash, she uses it, as if *cash burns* in her hands. A generalization of such policies to a setting with lumpy purchases is proposed by Arango-Arango et al. (2018), that present a model where the agent uses cash whenever they she has enough ($s \leq m$) and cards otherwise. The *cash burns* policy, if taken at face value, implies that cash is used whenever $s \leq m$: in other words, the share of *unforced card payments* is negligible. Figure 1b shows that this is not the case. Across all survey waves, a substantial fraction (27 to 49%) of unforced transactions are paid by card; such a “violation” of the simple *cash burns* policy appears to be getting increasingly common over time. Therefore, a theory in which cards are a residual means of payment, employed only when there are no other viable options, seems at odds with this evidence.

FIGURE 2: Share of card payments for different m , s and $m' = m - s$.



Note: The left panel displays the share of payments settled using cards for bins defined in terms of cash holdings at payment (m) and transaction size faced (s). Numbers denote the number of observations falling in each bin. We focus on transactions where m and s are smaller or equal than 100 euros to avoid having cells with a very small number of observations (different sets of transactions/cash holdings are considered in Appendix, see Figure A.3 and Figure A.4). The right panel displays the shares of households paying using cards for bins defined in terms of cash holdings remaining in case the agent settles the payment using cash (*implied residual cash holdings* $m' = m - s$). A nonparametric fit ($h = 10$) with 95% confidence intervals is overlaid to the plot. Only unconstrained transactions are considered, and transactions with $m = s$ are omitted. Data from ECB payment diaries: SUCH (2016) and SPACE (2019, 2021-22 and 2023-24).

2.3 More evidence on payment behavior

In this Section we document a new pattern concerning payment choices. In the left panel of Figure 2, we display how the payment method choice (cash vs cards) varies as a function of the *amount of cash* balances and the *size of the purchase*. The graph shows the proportion of transactions settled using cards, as opposed to cash, for different bins defined by cash holdings m and purchase size s . We focus on *unforced* payments, by excluding both the transactions in which cash is not a feasible payment method ($s \geq m$ or cash not being accepted) and those in which card payments are not available (the shop does not accept cards or the individual does not have access to a card). Focusing on unforced payments is crucial as it nets out the mechanical effect of m and s on payment choices through their impact on available payment options.¹⁴

The Figure reveals that, for a given level of cash holdings m , the probability of using

¹³See for example Stix (2004), Arango, Huynh, Fung, et al. (2012), Huynh, Schmidt-Dengler, and Stix (2014) and Arango, Huynh, and Sabetti (2015).

¹⁴Notice that both estimated effects are likely to be biased upwards when payment choice sets are not exactly observed, for two reasons. First, the probability of a card being accepted by merchants is increasing in the size of the transaction. In Figure A.2, we show that this phenomenon is evident in our sample as well. Second, when transactions are larger it is more likely to have insufficient cash to settle the purchase.

cards rises as the size of the transaction s increases. This result is consistent with previous empirical findings for the US and Canada (Klee (2008), Wang and Wolman (2016), Wakamori and Welte (2017)). Moreover, for a given purchase size s , individuals are less likely to use cards to settle a purchase when cash balances m are larger: this finding is in line with empirical evidence for several countries (Stix (2004), Arango, Huynh, Fung, et al. (2012), Huynh, Schmidt-Dengler, and Stix (2014), Arango, Huynh, and Sabetti (2015), Bagnall et al. (2016)).¹⁵

These facts offer strong evidence that both the transaction size s and the amount of cash holdings m affect individuals' payment choices. Questionnaire data available in the SUCH survey further corroborate this claim. Respondents were asked the question “Which of the following influences your decision to pay with cash or card or other non-cash payment methods?”. As shown in Figure A.5 in the Appendix, their responses align closely with our empirical findings based on observed behavior: the level of cash holdings and the size of the transaction are the most frequently cited factors, each mentioned by over 50% of respondents as key drivers of their payment method decisions. This evidence reinforces our argument that payment behavior is state-dependent. In particular, differences in payment method choices arise not only from time-invariant heterogeneity across consumers (e.g., demographics), but also – importantly – from within-consumer variation driven by fluctuations in m and s across transactions.¹⁶

The left panel of Figure 2 also illustrates that when the transaction size s is very close to m (in the graph, these transactions lie in the area just below the 45° line) cards are employed more often. We look at this phenomenon more closely in the right panel of the Figure, where we display the proportion of card payments for different levels of *implied residual cash holdings* (residual cash holdings in case of a cash payment, i.e., $m' = m - s$). It shows that the

¹⁵In principle, these results could be driven by selection: individuals who prefer to use cards might have on average little cash with them and purchase more expensive goods. To rule out this possibility, we run linear probability models in where we add a large set of demographic and transaction-specific controls; we also include random effects exploiting the panel dimension of our data. We find statistically significant relationships between the size of purchases/the level of cash balances and payment method choices (results are reported in Appendix A.3). On average, a 10 EUR increase in payment size s is associated with a 4.1pp (8%) increase in card usage probabilities, while a 10 EUR increase in cash holdings m is associated with a 0.6pp (2%) decrease in the probability of using cards.

¹⁶Among the respondents to SUCH/SPACE that report at least 2 transactions, $x\%$ used both cash and cards, suggesting that one should strongly reject the hypothesis that payment choices only vary across, but not within, consumers.

probability of card payments rises as $m' = m - s \rightarrow 0$. Households avoid using cash when doing so would lead to a near depletion of their money holdings, suggesting the existence of a precautionary motive to hold a buffer stock of cash: consumers anticipate that cash may be needed for transactions taking place in stores where payment cards cannot be used.¹⁷

3 An inventory model with a cash vs card choice

In this Section we present an inventory model of cash management with a choice between cash and card payments. We begin by describing the setup and the sequence problem faced by the agent. In [Section 3.1](#) we present the Bellman equation and characterize the optimal policy. In [Section 3.2](#) we compute the model-implied observable statistics, and in [Section 3.3](#) we compare the predictions of our model with previous theoretical and empirical work.

We consider an agent who finances an expenditure stream given by a compound Poisson process, whose cumulative value at time t is given by $e(t) = \sum_{i=1}^{N(t)} s_i$, where $s_i > 0$ is the size of the i -th purchase, captured by a (continuous) random iid variable with cdf $F(s)$, and $\{N(t) : t \geq 0\}$ is the counter associated to a Poisson process with rate λ . We make this assumption to model a realistic expenditure flow, with purchase opportunities arising at random times $\{T_i\}$ and transactions having discrete sizes $\{s_i\}$. Agents can pay for their purchases using either cash or a payment card.¹⁸ Using cards entails a fixed fee κ , independent of the value of the purchase and meant to capture both monetary fees and the non-pecuniary costs of using cards instead of cash, such as e.g., habit formation in payment behavior and privacy concerns. We do not take a stand on the sign of κ , allowing for $\kappa > 0$ to reflect the costs of card usage (monetary costs, privacy costs) or alternatively $\kappa < 0$ to reflect the benefits of a card payment (e.g. safety, convenience).¹⁹ We assume that cards are accepted in a fraction

¹⁷[Eschelbach and Schmidt \(2013\)](#) find a similar pattern in German payment diary data, attributing it to precautionary behavior by consumers who want to avoid using the entire amount of cash in their wallets, as they might need it for future purchases. In [Appendix A.3](#) we present regression output from linear probability models (adding a full set of controls to take care of selection) that corroborates this claim: we show that the positive effect of an increase in s on the probability to use cards becomes larger in magnitude as m decreases, consistently with individuals timing their use of card payment to avoid settling with cash those large transactions that would drain their balances. We corroborate our interpretation that imperfect card acceptance by merchants is an important driver of such precautionary motive in [Appendix A.4](#).

¹⁸In the model we use the word *card* to denote any cashless payment option, as we did in [Section 2](#). Even though each non-cash payment method has different properties (e.g. debit vs credit cards), our model abstracts from such differences.

¹⁹Using shopping diary data, [van der Cruysen, Hernandez, and Jonker \(2017\)](#) show that the habit of paying

$\phi \in (0, 1)$ of stores, which is exogenous and independent of the value of the purchase.

Let m denote the level of cash balances. Holding m units of cash entails an opportunity cost Rm , while adjusting cash holdings through a cash withdrawal (w) has a fixed cost $b > 0$. Cash balances must be larger than the size of the transaction in order to settle a purchase with cash ($m > s$). When cash on hand is insufficient to carry out a purchase and the store does not accept cards, the agent cannot consume: we assume that this event entails a cost u , independent of the size of the purchase missed.²⁰ Letting ρ be the discount rate, the agent's problem is described by

$$\begin{aligned}
v(m) = \min_{\{(\tau_j, w_j)\}_{j=0}^{\infty}, \{\alpha_i\}_{i=0}^{\infty}} \mathbb{E} & \left(\int_0^{\infty} e^{-\rho t} R m(t) dt + \sum_{j=0}^{\infty} e^{-\rho \tau_j} b \right. \\
& \left. + \sum_{i=0}^{\infty} e^{-\rho T_i} \left(\kappa \cdot \mathbf{1}(\alpha_i = \text{card}) + u \cdot \mathbf{1}(\alpha_i = \text{lose}) \right) \right) \Big| m(0) = m \\
\text{subject to } m(t) = m(0) & + \sum_{\tau_j \leq t} w_j - \sum_{T_i \leq t} s_i \cdot \mathbf{1}(\alpha_i = \text{cash})
\end{aligned} \tag{1}$$

where $v(m)$ is the value function representing the expected discounted costs as a function of current cash holdings m . The agent controls the sequence of withdrawal times $\{\tau_j\}_{j=0}^{\infty}$ and withdrawal amounts $\{w_j\}_{j=0}^{\infty}$, as well as the payment method decisions $\{\alpha_i\}_{i=0}^{\infty}$ at each purchase arrival time T_i , when the agent chooses $\alpha_i \in \{\text{cash}, \text{card}, \text{lose}\}$: paying with cash (if $m \geq s_i$), paying with card at cost κ , or losing the purchase at cost u . Cash withdrawals incur a fixed cost b . The constraint describes how cash-withdrawals w_i and cash-expenditures s_i affect the evolution of $m(t)$.

with cash is a major driver of consumers' payment behavior at the point of sale in the Netherlands. Drawing on data from the 2021-22 SPACE survey questionnaire, [Nocciola and Zamora-Peréz \(2024\)](#) find that privacy concerns, the speed of settlement, and budget control features are major drivers of preferences for cash. In principle, it could instead be that using cards entails a fixed *benefit* with respect to using cash ($\kappa < 0$), as the former are perceived as safer ([Schuh and Stavins \(2016\)](#)), easier to use and are possibly more convenient (due to loyalty rewards and cashback programs, see [Ching and Hayashi \(2010\)](#)).

²⁰We assume that u is large enough that the agent always prefers to complete transactions if she can do so.

3.1 A characterization of the optimal payment policy

Consider an agent with cash balances m . Define

$$m^* = \arg \min_m v(m), \quad (2)$$

and let $v^* = v(m^*)$. The value function obeys the Bellman equation

$$\begin{aligned} \rho v(m) = \min \bigg\{ & \rho(v^* + b) \text{ , } Rm + \lambda(1 - F(m))(\phi\kappa + (1 - \phi)u) + \\ & + \lambda \int_0^m \left(\phi \min \{v(m - s) - v(m), \kappa\} + (1 - \phi)(v(m - s) - v(m)) \right) dF(s) \bigg\}. \end{aligned} \quad (3)$$

The outer min operator describes the agent's choice between paying the cost b and refilling her wallet, versus choosing not to withdraw cash. In the latter case, with probability $1 - F(m)$ the incoming purchase is too large to be paid in cash, namely $s \geq m$: if the store accepts cards, which happens with probability ϕ , a cost κ is paid and a card payment occurs. Otherwise the purchase is lost and the agent incurs a loss of size u . When the size of the incoming purchase is smaller than m and the store accepts cards, the agent faces a nontrivial payment choice, represented by the inner min operator. If she uses cash, the state jumps to $m - s$. Alternatively, the agent pays the fixed cost κ , uses the card and the cash balance is unchanged. When the store does not accept cards, cash is used and the value function changes from $v(m)$ to $v(m - s)$.

Next, we discuss the agent's optimal policy. This is given by two decision rules: a *withdrawal policy*, determining whether (and how much) to withdraw for any value of current cash holdings m , and a *payment policy*, which determines whether to use cash or cards to settle an expenditure of size s when having $m \geq s$ units of cash at hand.

Withdrawal policy. We make the following simplifying assumption that holds under a wide range of parameters and enables us to provide a sharper description of the model's solution:²¹

²¹The model setup is close to the stochastic, continuous-review inventory model for the case of compound Poisson demand, for which the optimal policy belongs to the class of impulse-control trigger-target policies $(\underline{m}, m^*)$ often called Ss policies, see [Perera and Sethi \(2023\)](#). The lack of a diffusion or drift component makes the continuous-time problem mathematically close to a discrete-time one, as it can only be optimal to adjust after a negative jump in the state occurs. Our model departs from the standard setup as the agent has the possibility to finance part of the consumption stream using cards. As the optimality of trigger-target policies in such a setting has not been proven, we check their optimality in our context numerically. Numerical experiments suggest that

Assumption 1. *There exists $\underline{m} \geq 0$ such that $v(m) = v^* + b$ for any $m \leq \underline{m}$, and $v(m)$ is continuous and strictly decreasing on $[0, m^*]$.*

[Assumption 1](#) posits that the optimal withdrawal policy is of the Ss form: it is characterized by a trigger value $\underline{m}$ and a target value m^* , such that a cash-withdrawal takes place if $m \leq \underline{m}$, and the cash balance is reset to $m^* > \underline{m}$. The value function v is flat in the withdrawal region, namely $v(m) = v^* + b$ for $m \leq \underline{m}$. This is the highest level that the value function takes, i.e. the most costly position the agent experiences. The value function is (weakly) decreasing over $(\underline{m}, m^*)$. In [Figure 3a](#) we display the value function $v(m)$ and the optimal withdrawal policy, described by $\underline{m}$ and m^* . We also define the value $\tilde{m}$, which is the level of cash holdings that satisfies $v(\tilde{m}) = v^* + b - \kappa$. It is immediate that the card will not be used if $m < \tilde{m}$, since using the card would give a value higher than $v^* + b$. Notice that if $\kappa > 0$ then $\underline{m} < \tilde{m}$, as discussed below. [Figure 3a](#) shows that the optimal policy features a positive trigger value $\underline{m} > 0$, i.e., the agent withdraws cash before emptying her wallet. This happens because the model features a precautionary motive for holding cash: the agent needs to insure against shopping trips in which merchants might not accept cards (which might entail a cost u).

Payment policy. Under [Assumption 1](#), it is possible to characterize the payment policy in detail. Let $p(m, s) = \{0, 1\}$ denote which payment method is employed when facing a purchase of size s and having m units of cash balances on hand, conditional on cards being accepted, with $p(m, s) = 1$ denoting the choice of the card payment. For $s \leq m$, we have

$$\begin{aligned} p(m, s) &= \arg \min_{p \in \{0, 1\}} \mathbb{1}(p = 0) (v(m - s) - v(m)) + \mathbb{1}(p = 1) \kappa \\ &= \begin{cases} 0 & \text{if } v(m - s) - v(m) \leq \kappa \quad \text{cash is used,} \\ 1 & \text{if } v(m - s) - v(m) > \kappa, \quad \text{card is used.} \end{cases} \end{aligned} \quad (4)$$

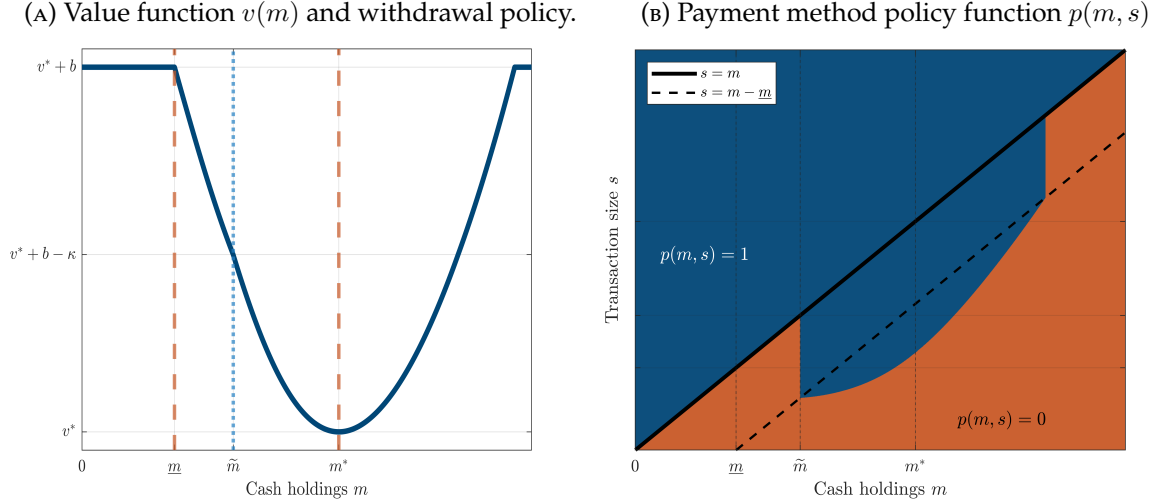
Feasibility requires $p(m, s) = 1$ if $s > m$. We establish the following:

Lemma 1. *Let [Assumption 1](#) hold. If $p(m, s) = 0$, then $p(m, s') = 0$ for any $s' < s$. If $p(m, s) = 1$, then $p(m, s'') = 1$ for any $s'' > s$.*

Proof. See [Appendix B.1](#). ■

unless the pdf f associated with F has mass points or “holes” in its support, the optimal policy is of the Ss type.

FIGURE 3: Optimal policy.



Note: The right panel shows the value function $v(m)$ that solves problem (3). The right panel shows the policy function, with $p(m, s) = 1$ representing a card payment. The thick black line depicts the 45 degree line. The dashed black lines depicts the combination of levels of cash holdings m and purchase sizes s for which $m - s = \underline{m}$: above this line, if the agent pays cash she immediately withdraws afterwards. The parameters are those used in the calibration of the model for 2021-22 (see Section 4).

Lemma 1 establishes that payment method choices are monotone in the size of the transaction, for any level of cash holdings m . When facing a payment method decision, the agent compares the cost of paying with cash, i.e., the instantaneous change in the value of the problem $v(m - s) - v(m)$, with the cost of using cards κ . If v is weakly decreasing, the change in the value of the problem will always get more costly as s increases, for any m .

Note that any transaction s satisfying $m - s \in [0, \underline{m})$ is big enough that, if the agent decides to settle it using cash, she will need to visit an ATM immediately after. We label such purchases as *large*, as they trigger a cash withdrawal if settled in cash. The next lemma shows that there exists a critical threshold for cash holdings, $\tilde{m}$, such that large transactions are settled with the card only if $m > \tilde{m}$. This results illustrates the interactions that arise between the payment method choice and cash holdings:

Lemma 2. Let [Assumption 1](#) hold. Then, if $0 < \kappa < b$, there exists $\tilde{m} \in (\underline{m}, m^*)$ satisfying

$$v(\tilde{m}) = v^* + b - \kappa, \quad (5)$$

such that whenever $m - s \in [0, \underline{m})$,

$$p(m, s) = \begin{cases} 0 & \text{if } m \in [\underline{m}, \tilde{m}], \quad (\text{cash is used}) \\ 1 & \text{if } m \in (\tilde{m}, m^*]. \quad (\text{card is used}) \end{cases} \quad (6)$$

Proof. See [Appendix B.2](#). ■

Lemma 2 states that if $0 < \kappa < b$, when facing a large purchase (i.e., such that $m - s \in [0, \underline{m})$), the agent adopts the following behavior. If the current cash balances are small, the difference between the cost of using the card ($v(m) + \kappa$) and the cost of using cash and withdrawing ($v^* + b$) is large, so that individuals prefer to use cash. The intuition is that when m is below $\tilde{m}$ and a large purchase arises, it is not worth it to pay κ to keep cash holdings at the current level, which is far from the optimal one m^* . If a large purchase occurs when m is above $\tilde{m}$, instead, the agent finds it optimal to pay κ to keep her cash holdings unchanged.

Combining the above Lemmas yields a Proposition that characterizes payment choices under the impulse control policy with trigger $\underline{m}$ and target m^* :

Proposition 1. Let [Assumption 1](#) hold. Then, for any $0 < \phi \leq 1$, we have that

1. for $\kappa \leq 0$, we have that $p(m, s) = 1$ (card is used) for all $m \in [\underline{m}, m^*]$ and for all s ;
2. for $0 < \kappa < b$, there exists $\tilde{m} \in (\underline{m}, m^*)$ such that
 - (a) for any $m \in [\underline{m}, \tilde{m}]$, $p(m, s) = 0$ (cash is used) for all $s \leq m$;
 - (b) for any $m \in [\tilde{m}, m^*]$, there exists a transaction size $\underline{s}(m) \leq m - \underline{m}$, satisfying

$$v(m - \underline{s}(m)) = v(m) + \kappa, \quad (7)$$

such that $p(m, s) = 0$ (use cash) for all $s < \underline{s}(m)$, while $p(m, s) = 1$ (use card) for all $s \geq \underline{s}(m)$.

3. for $\kappa \geq b$, we have that $p(m, s) = 0$ (use cash) for all $m \in [\underline{m}, m^*]$ and $s \leq m$.

Proof. See [Appendix B.3](#). ■

[Proposition 1](#) highlights a number of features of payment choices. The payment policy for cases 1 and 3 is easily explained: when $k \leq 0$, using the card is cheaper than using cash, therefore the card option dominates cash. When $\kappa > b$, on the other hand, it is never optimal to use cards as using cash is more convenient, even when doing so leads to a shortfall and to the associated withdrawal; in this case, cards are a residual means of payment used only for forced transactions (when $s > m$). The payment policy for the case in which $0 < \kappa < b$ is depicted in [Figure 3b](#). The main result is that using cards when cash on hand is sufficient to carry out the incoming transaction (i.e., below the 45 degree line) is optimal as long as the purchase size is larger than the threshold $\underline{s}(m)$, which varies with the level of cash holdings, m . In particular, when cash balances are sufficiently large ($m \geq \tilde{m}$), the agent always uses cards for all purchases that would lead to a withdrawal if paid in cash (i.e., such that $m - s \leq \underline{m}$).²² The reason is that when $\kappa < b$ the agent is willing to pay κ in order to avoid paying b , provided the current level of cash holdings is sufficiently close to the optimal one (i.e., it is larger than $\tilde{m}$). As shown by the blue area below the dashed line in [Figure 3b](#), it may also be optimal to use cards for some transactions that would not lead to an immediate withdrawal but would still push the agent too close to $\underline{m}$ (i.e., such that $m - s \in (\underline{m}, m - \underline{s}(m)]$), to avoid visiting a region of the state space where the probability of losing a purchase becomes non-negligible. Moreover, cash is always employed for sufficiently small transaction sizes, s (given that using cards entails a fixed cost). When cash balances are small ($m < \tilde{m}$), cards are not used. In such cases, the agent is so far from the optimal level of cash holdings m^* that it is never worth to pay the fixed cost to keep the cash balances unchanged. Note that [Proposition 1](#) holds true for any level of card acceptance $\phi \in (0, 1]$:²³ even though changes in ϕ affect the values of $\underline{m}$, m^* and $\tilde{m}$, the features of the payment policy described in the Proposition and portrayed in [Figure 3b](#) remain valid. Finally, we note that even when $\phi = 1$ the agent finds it optimal to use cash on some occasions, as long as $\kappa > 0$.

²²This follows from (i) $\underline{s}(m) \leq m - \underline{m}$ and (ii) $p(m, s) = 1$ for all $s \geq \underline{s}(m)$, which imply that $p(m, s) = 1$ whenever $s \geq m - \underline{m}$, i.e., whenever $m - s \leq \underline{m}$.

²³The proposition holds vacuously for $\phi = 0$ as the function p is not defined if cards are never accepted.

3.2 Model-implied observable statistics

We use the policy $(\underline{m}, m^*, p)$ to compute a set of moments that describe the cash management and payment method decisions of households, and that can be compared to observed statistics.

Cash management statistics. Average cash balances M in the economy can be obtained by computing the expected value of m under the invariant distribution of cash holdings $h(m)$. Notice that, due to the absence of a drift component, the density has a mass point at m^* . With a slight abuse of notation, we denote with $h(m)$ for $m \in [\underline{m}, m^*)$ the density of m , integrating to $1 - h(m^*)$, where $h(m^*) = \mathbb{Pr}(m = m^*)$. We show in [Appendix B.5](#) that for any $m \in [\underline{m}, m^*)$ the invariant distribution satisfies the KFE

$$h(m) = \frac{\int_{\underline{m}}^{m^*} h(m') f(m' - m) (1 - \phi p(m', m' - m)) dm' + h(m^*) f(m^* - m) (1 - \phi p(m^*, m^* - m))}{F(m) - \int_0^m f(s) \phi p(m, s) ds}, \quad (8)$$

with boundary condition $\int_{\underline{m}}^{m^*} h(m) dm + h(m^*) = 1$. We solve (8) numerically and obtain the invariant distribution h ; we can then trivially compute average cash holdings $M = \int_{\underline{m}}^{m^*} m h(m) dm + m^* h(m^*)$, as well as median cash holdings. We also want to compute the number of withdrawals per unit of time. To do that, we need to compute the function $\mathcal{T}(m)$, which gives the expected amount of time until the next withdrawal when cash balances are equal to m . We show in [Appendix B.6](#) that $\mathcal{T}(m)$ obeys the functional equation

$$\mathcal{T}(m) = \frac{1 + \lambda \int_0^{m - \underline{m}} f(s) (1 - \phi p(m, s)) \mathcal{T}(m - s) ds}{\lambda \left[F(m) - \int_0^m f(s) \phi p(m, s) ds \right]}. \quad (9)$$

After solving for $\mathcal{T}$, it is possible to obtain the average number of withdrawals performed by the agent per unit of time via the fundamental theorem of renewal theory $n = 1/\mathcal{T}(m^*)$, i.e., by exploiting the fact that the average number of withdrawals is the reciprocal of the length of a withdrawal cycle. It is also possible to compute the average cash on hand when a withdrawal takes place, denoted by $\underline{M}$, as well as the average size of withdrawals W (see [Appendix B.7](#) for details).

Realized expenditure. Since some purchase opportunities are lost, the actual expenditure per unit of time e is weakly smaller than the intended expenditure $\tilde{e} = \lambda \int_0^\infty sf(s)ds$. In particular, actual expenditure is given by

$$e = \lambda \cdot \left[\int_0^{\underline{m}} sf(s)ds + \int_{\underline{m}}^\infty sf(s) (1 - H(s)(1 - \phi)) ds \right],$$

where the very last term is the probability of losing a purchase opportunity when cash balances are lower than z and the visited shop does not accept cards. The number of “realized” purchases is

$$\hat{\lambda} = \lambda \left(\int_{\underline{m}}^{m^*} h(m) (F(m) + (1 - F(m)) \phi) dm + h(m^*) (F(m^*) + (1 - F(m^*)) \phi) \right).$$

Payment choice statistics. The card share of expenditures γ is given by

$$\gamma = \frac{\lambda \phi \left(\int_{\underline{m}}^{m^*} h(m) \gamma(m) dm + h(m^*) \gamma(m^*) \right)}{e}, \quad (10)$$

where $\gamma(m) = \int_0^m sf(s)p(m, s)ds + \int_m^{+\infty} sf(s)ds$ is the share of expenditure paid with cards when having m units of cash on hand. We also want to capture how often cards are used conditional on having both options available, i.e., for *unforced* purchases. The card share of unforced expenditure $\tilde{\gamma}$ is computed as

$$\tilde{\gamma} = \frac{\lambda \phi \left(\int_{\underline{m}}^{m^*} h(m) \left(\int_0^m sf(s)p(m, s)ds \right) dm + h(m^*) \left(\int_0^{m^*} sf(s)p(m^*, s)ds \right) \right)}{\lambda \phi \left(\int_{\underline{m}}^{m^*} h(m) \left(\int_0^m sf(s)ds \right) dm + h(m^*) \left(\int_0^{m^*} sf(s)ds \right) \right)}. \quad (11)$$

It is also possible to compute the share of *purchases*, and not expenditure, settled using cards, both overall and when having both options available: we denote these two statistics by γ_n and $\tilde{\gamma}_n$ (see [Appendix B.7](#) for details).

3.3 Relationship with previous work

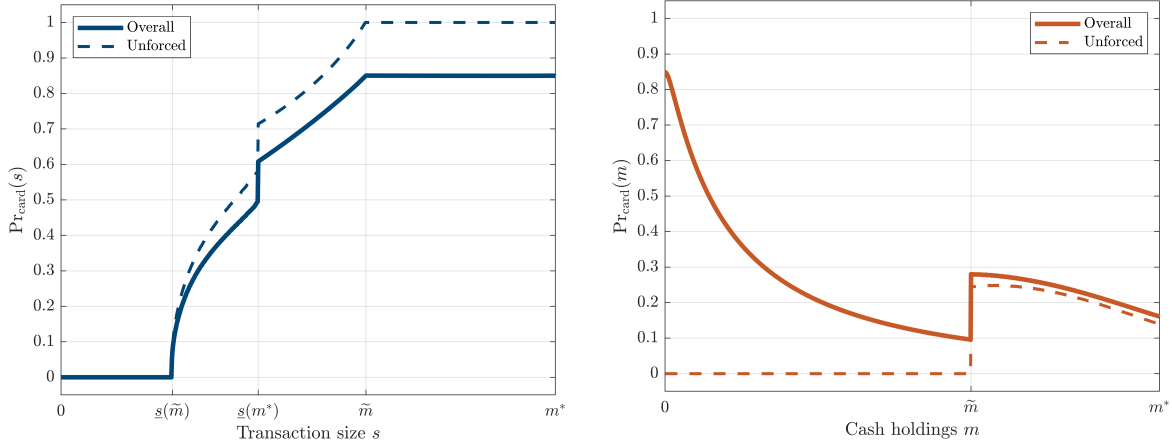
Theoretical models. Our model features a precautionary motive for holding cash which (if sufficiently strong) induces the agent to withdraw when she still has cash on hand ($\underline{m} > 0$). In particular, individuals want to hedge against an excessive frequency of card payments, which are costly to perform, and against the risk of missing a purchase. When m is close to zero, indeed, it is highly likely that the incoming transaction is too large to be settled with cash, which results in a forced card payment or, if the merchant does not accept cards, in a missed purchase. A similar feature appears in the model with uncertain lumpy expenditures by [Alvarez and Lippi \(2013\)](#), even though in their case the agent can finance lumpy transactions larger than her current cash balances by withdrawing *before* the purchase takes place, something that we do not allow here.

As for the payment method choice, our model can be seen as a generalization of those in [Whitesell \(1989\)](#) and [Alvarez and Lippi \(2017\)](#). In [Whitesell \(1989\)](#), households follow a transaction-size threshold policy, paying for large purchases (larger than $\underline{s}$) with cards and for small ones with cash. Here they adopt a similar behavior, even though the critical level of the transaction-size depends on the cash at hand, given by the function $\underline{s}(m)$. In [Alvarez and Lippi \(2017\)](#), households always use cash to settle transactions when they have enough. In their model, the expenditure stream is infinitesimal, hence we can compare their predictions with those we obtain for very small transactions. In our model too, when s is close to zero, it is optimal to use cash. This holds since using cash requires no immediate costs, and the value of the problem is only slightly affected by a tiny cash purchases, i.e., $m' = m - s \sim m$.

Empirical evidence. The optimal policies and the observable moments produced by our framework are broadly consistent with existing empirical evidence on payment choices and cash management, as well as with the new facts on payment behavior highlighted in [Section 2](#). First, as shown in the left panel of [Figure 4](#), the solution of our model implies that²⁴ the probability to use cards is increasing in the size of the transaction faced, as documented by

²⁴After solving for the payment choice policy function $p(m, s)$ and for the invariant distribution $h(m)$, it is possible to compute the probability that cards are used as a function of s (the size of the transaction), m (the amount of cash on hand) and $m' = m - s$ (the amount of implied residual cash holdings). Details are provided in [Appendix B.8](#).

FIGURE 4: Probability of card usage as a function of s and m .



Note: In the left panel, we display the model-implied predicted probability of using cards as a function of the size of the incoming transaction s . In the right panel, we display the same probability as a function of the size of the incoming transaction s . Solid lines represent raw probabilities, while dashed ones represent the probability of using cards conditional on both payment methods being available. Parameters are those obtained from the calibration of the model for 2021-22 (see Section 4).

Klee (2008) and many others.²⁵ Second, consistently with the empirical literature showing that the probability to use cash increases as cash balances get larger (Stix (2004), Huynh, Schmidt-Dengler, and Stix (2014), Arango, Bouhdaoui, et al. (2014)), we show in the right panel of Figure 4 that for $m > \tilde{m}$ this pattern emerges in our model as well. A unique prediction of our model is that the intensity of cash usage (when both methods are available) is U-shaped in m . Not only the agent uses cash more often when she has a lot, but she does it often even when she has very little, as she knows that she will need to withdraw soon.²⁶ Overall, our payment choice rule closely resembles the patterns observed in the data, as can be seen by comparing the left panel of Figure 2, where we display the intensity of card usage as a function of m and s in the data, and Figure 3b, where we plot its model counterpart: qualitatively, the two plots look very similar. Third, our model also qualitatively reproduces the feature documented in the right panel of Figure 2, i.e., that the probability of using cards as a function of implied remaining cash holdings $m' = m - s$ (what would be left in the

²⁵As shown in the left panel of Figure 4, cash is always used for purchases of size smaller than $\underline{s}(\tilde{m})$. Cards are used more frequently as s rises, with a discontinuity at $\underline{s}(m^*)$ due to the distribution of cash holdings having a mass point at m^* . The probability of card usage reaches its peak for $s \geq \tilde{m}$, where it reaches ϕ (the card acceptance rate): for purchases this large, cards are always used when they are accepted.

²⁶The right panel of Figure 4 shows that the probability of using cards for unforced purchases as a function of m peaks at $\tilde{m}$ and then falls. This happens because the closer m is to $\tilde{m}$, the steeper is the value function, and so the larger is the set of transactions that make the value of the problem increase by enough that card usage is warranted.

agent's wallet if the incoming purchase is settled using cash) is flat for large enough m' and steeply rises as m' approaches zero.²⁷ Finally, we note that the model reproduces the empirical relationship between card acceptance and cash management: as ϕ falls, the precautionary motive for holding a cash buffer gets stronger, and the level of cash holdings $\underline{m}$ that triggers a withdrawal rises.²⁸

4 Mapping the model to the data

In this Section, we calibrate the model mapping it to the data from payment diaries collected in the third wave of the ECB SPACE survey (2023-2024). [Section 4.1](#) describes our calibration strategy in detail. [Section 4.2](#) presents the results on the model's parametrization.

4.1 Calibration strategy

The set of parameters is given by $\{\rho, \phi, F, R, b, \kappa, u, \lambda\}$. We calibrate the model at the yearly frequency, using a mixture of external information for parameters that can be normalized or directly identified from the data and internal calibration (through minimum distance) for the remaining parameters.

Externally calibrated parameters. The discount rate is set to $\rho = 0.05$ to obtain an annual discount factor $\beta = 1/(1 + \rho) \simeq 0.95$. The card acceptance rate ϕ is set equal to the share of merchants $\hat{\phi}$ who do not accept cards as a payment method. Given that imperfect acceptance may lead to some purchases being lost, we do not directly set the number of purchase opportunities λ equal to the observed number of purchases per year $\hat{\lambda}$ - on the contrary, we estimate λ internally, as explained below. We target a total *realized* expenditure of $e = \hat{\lambda} \int_0^{+\infty} sf(s)ds = \hat{\lambda} \mathbb{E}(s) = 365$, so that the average daily expenditure is equal to one. We calibrate the size distribution of payments from the data, using the following procedure. First, we assume that the distribution of transaction sizes F is lognormal, i.e., that $s \sim LN(\mu_s, \sigma_s^2)$.²⁹ Our normalization of yearly expenditure delivers a parametric restriction on

²⁷See [Figure D.1](#) in [Appendix D](#), where we plot the model-implied function $\text{Pr}_{\text{card}}(m')$.

²⁸See [Figure D.2](#) in [Appendix D](#), where we show how the value function $v(m)$ and the withdrawal policy $(\underline{m}, m^*)$ change as ϕ rises from the levels observed in 2021-22 to that prevailing in 2023-24 (a 4pp increase).

²⁹This is consistent with [Figure A.7](#), where we display the distribution of log payment sizes for all three surveys, which closely resemble a normal distribution.

TABLE 2: Calibration results and model fit.

Calibration results		Model fit	Data (2023-24)	Model
<i>Externally calibrated parameters</i>		<i>Targeted moments</i>		
Size distribution F , location μ_s	-1.95	Cash balances, M/e	1.34	1.30
Size distribution F , scale σ_s^2	2.63	N. cash withdrawals per year, n	96.40	96.68
Card acceptance rate ϕ	0.89	Card share of unforced expenditure $\tilde{\gamma}$	0.46	0.46
		N. purchases per day $\hat{\lambda}/365$	1.90	1.91
<i>Internally calibrated parameters (minimum distance)</i>		<i>Untargeted moments</i>		
Withdrawal cost b (EUR)	0.003	Cash balances, M/e (median)	0.95	1.32
Card usage cost κ (EUR)	0.002	Cash balances, M/c	3.22	5.38
Lost purchase cost u (EUR)	0.976	Withdrawal size, W/M	1.18	0.70
Purchase oppurt. per day $\lambda/365$	1.92	Card share of expenditure γ	0.58	0.76

Note: This table contains our calibration results and information on the fit of the model to observed moments (SPACE data, wave 3, 2023-24). The parameter σ_s^2 is calibrated to match the coefficient of variation of purchase sizes $\widehat{\mathbb{C}\mathbb{V}}_s$, and μ_s results from a normalization of total yearly expenditure to 365. Opportunity costs are normalized to $R = 10\%$. Cash balances are normalized by the overall daily expenditure e ; we also display cash balances divided by daily *cash* expenditures c .

F , i.e., $\mathbb{E}(s) = \exp\left(\mu_s + \frac{\sigma_s^2}{2}\right) = 365/\hat{\lambda}$. Targeting the empirical coefficient of variation of transaction sizes $\widehat{\mathbb{C}\mathbb{V}}_s$, we obtain another parametric restriction, i.e., $\sigma_s^2 = \log\left(\widehat{\mathbb{C}\mathbb{V}}_s^2 + 1\right)$. The observed frequency of purchases and the coefficient of variation of payment sizes allow us to recover (μ_s, σ_s) , and henceforth the cdf and the pdf f . The values of the externally calibrated parameters are displayed in [Table 2](#).

Internally calibrated parameters. We estimate the remaining five parameters $\theta = \{b, R, \kappa, \lambda, u\}$ in the following way.³⁰ We set $R = 10\%$, which we deem as a reasonable upper bound for the yearly opportunity cost of holding cash.³¹ We then estimate the parameter vector $\tilde{\theta} = \{b, \kappa, u, \lambda\}$ in order to match four moments that summarize the cash management and payment behavior of the euro area households. The targeted moments are i) the average cash balances relative to daily expenditure M , ii) the number of withdrawals per year n , iii) the share of expenditure settled using a card when both options are available $\tilde{\gamma}$, and iv) the number of realized purchases per year $\hat{\lambda}$.

The average cash balance M informs us on the opportunity cost of holding cash R , as the agent holds lower balances when the opportunity cost is higher. The frequency of

³⁰We note that the opportunity cost R is not identified separately from the other parameters. Let Θ be the space of feasible structural parameter vectors $\theta = \{b, R, \kappa, \lambda, u\}$, and let μ^θ be the set of model-implied observables. Consider $\theta' = \{Sb, SR, S\kappa, \lambda, Su\}$, for some $S \in \mathbb{R}_{++}$. Then $\mu^{\theta'} = \mu^\theta$.

³¹In [Knümann, Krüger, and Seitz \(2024\)](#), R is estimated to be around 7% in Germany, the largest economy in the euro area. We could not find any comparable estimate of R for other euro area countries.

withdrawals n is informative about the utility cost u of missing a purchase: the higher is u , the more often the agent will visit ATMs to make sure that she does not hang around with little cash on hand. The share of expenditure (voluntarily) settled with cards helps us to pin down the card fee κ (recall that, as shown in [Proposition 1](#), the share of expenditure voluntarily settled with cards approaches zero as $\kappa \rightarrow b$). We use $\hat{\lambda}$, the *observed* frequency of purchases, in order to recover λ , the frequency of purchase opportunities (see [Section 3.2](#)). We need to make sure that given the optimal policy and the size distribution of payments $F(s)$, the agent completes $\hat{\lambda}$ payments per year. We estimate $\tilde{\theta}$ via minimum distance,³² i.e., for a given set of empirical moments $\hat{\mu} = \{\hat{\mu}_i\}_{i=1}^4$, we choose $\tilde{\theta}^*$ such that $\tilde{\theta}^* = \arg \min_{\tilde{\theta}} \sum_{i=1}^4 \left(\frac{\mu_i^{\tilde{\theta}} - \hat{\mu}_i}{\hat{\mu}_i} \right)^2$, where $\{\mu_i^{\tilde{\theta}}\}_{i=1}^4$ are model-implied moments.

4.2 Calibration results

In [Table 2](#) we display calibrated values of the four structural parameters of the model $\{b, \kappa, u, \lambda\}$. We estimate a very small withdrawal cost b of 0.003 EUR. The fixed cost of using cards κ is even smaller than the cost of withdrawals b , as expected since households settle a sizeable share of unforced purchases with cards ($\tilde{\gamma} = 0.46$); we estimate a value of κ/b around 60%. We also estimate that around 2% of purchase opportunities are missed due to a combination of insufficient cash holdings when the opportunity to buy arises and imperfect card acceptance. Finally, we estimate a utility cost stemming from each missed purchase opportunity of around 1 EUR.

In the second part of [Table 2](#) we illustrate the fit of the calibrated model, both in terms of targeted moments and in terms of untargeted statistics. The model fits almost perfectly the four targeted moments. In terms of untargeted moments, we obtain model-implied values which are qualitatively in line with the data, even though the fit is not as good. For instance, we overestimate γ , the unconditional share of card purchases, by 18 percentage points. Since $c = (1 - \gamma)e$, this also implies that M/c is around 70% larger than its empirical counterpart. This shortcoming is likely due to the very simple nature of the model, which abstracts from many features of the data. For instance, in our model the probability of cards being accepted

³²We numerically implement a global search method on a Sobol quasi-random low-discrepancy grid of candidate parameter vectors, paired with a local search algorithm around the candidates for a global minimum.

ϕ is independent of the size of the purchase s , while in reality cards are more often accepted for larger transactions than for small-sized ones (see [Figure A.2](#)).

4.3 Welfare analysis

Let $\mathcal{C}$ denote the annual household's *cost of managing consumption transactions*. In a steady state this is

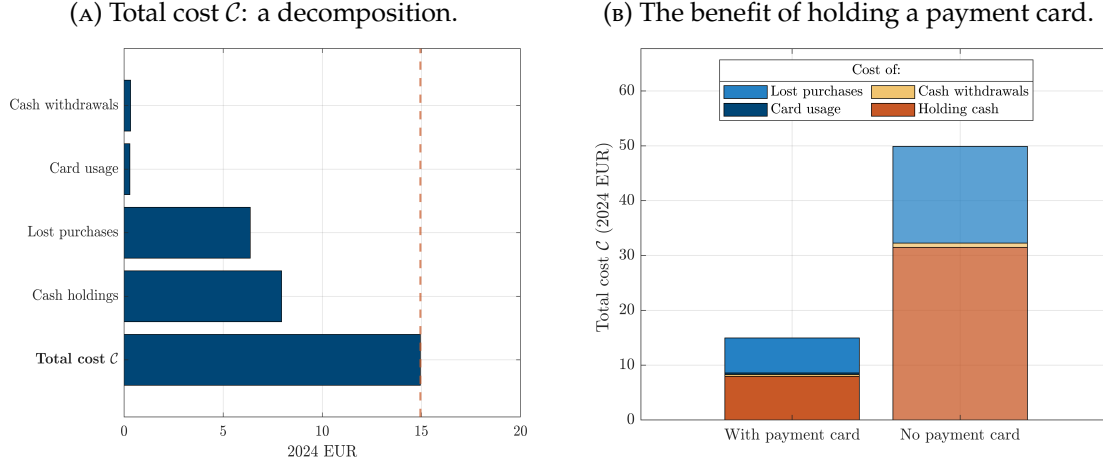
$$\mathcal{C} = RM + bn + \kappa\gamma_n\hat{\lambda} + u(\lambda - \hat{\lambda}), \quad (12)$$

where $\gamma_n\hat{\lambda}$ is the number of card purchases and $(\lambda - \hat{\lambda})$ is the number of missed purchase opportunities. As (12) shows there are four sources of costs in the model, namely i) the *opportunity costs* of holding cash (RM), ii) the cost related to *cash withdrawals* (bn), iii) *card usage costs* ($\kappa\gamma_n\hat{\lambda}$), and iv) the cost stemming from *lost purchases*, $u(\lambda - \hat{\lambda})$. Recall that we expressed cash holdings in terms of days of cash-expenditures, so that the cost $\mathcal{C}$ is also measured in days of consumption expenditures. We can thus express this value in 2024 euros by multiplying it by the average daily expenditure (approximately 61 euros). [Figure 5a](#) shows the estimated annual cost of managing consumption transactions, and how each of the four components contributes to the total cost: we estimate a cost of managing consumption transactions for the average household of approximately 15 euros per year, a remarkably low amount which suggests that the payment system faced by households imposes a relatively small burden on their budgets.

We note that the cost of transactions deliberately ignores that in order to pay for goods and services with either with cards or cash, households must hold liquid assets (typically overnight deposits), whose compensation is typically lower than that of alternative assets. We omit such costs because they are invariant with respect to the mix of cash vs card payment used, which is the core object of our exercise.

Heterogeneity. We use the richness of our data to estimate the model over different time periods and different demographic groups. In [Figure D.3](#) in the Appendix we display the estimated cost of managing consumption transactions, and its component, for each survey wave, from 2015-16 to 2023-24. Until 2019, the estimated total cost was around 20 euros per year, while it stabilised around 15 euros from 2021-22 onward. The difference is mostly due

FIGURE 5: The cost of managing consumption transactions



Note: The left panel shows a decomposition of the total cost of managing consumption transactions $\mathcal{C}$ (expressed in 2024 euros) for the estimated model. The right panel compares the total cost of managing consumption transactions $\mathcal{C}$ for the estimated model with that obtained in an alternative scenario where we set $\phi = 0$, i.e., where the agent cannot use her payment card.

to households' holding less cash in the post-pandemic period. We also estimate the model on 2023-24 data for different age groups, levels of completed education, income brackets and by gender.³³ Given that these groups have very different overall daily expenditures, and that total costs rise with consumption, in Figure D.4 in the Appendix we display the estimated total annual cost as a fraction of daily expenditure for each subgroup. The main result is that each of these demographic subgroups faces a very similar cost, as a proportion of their expenditure.

5 Three counterfactual experiments

In this Section, we use the estimated model to run three counterfactual experiments. Section 5.1 quantifies the welfare benefit of holding a payment card. Section 5.2 describes a counterfactual scenario in which card acceptance is near-universal ($\phi = .99$). Section 5.3 evaluates under which conditions the agent entirely abandons cash and a purely cashless economy emerges.

³³In Table 4 we display an extensive set of summary statistics for each wave of the survey. Table 6 presents our calibration for different time periods. Tables with summary statistics, estimated parameters and the model fit for all the other subsamples (by age, education and so on) are available upon request.

5.1 The benefit of holding a payment card

We use the estimated model to compute the benefit of holding a payment card. To do that, we solve the model again assuming that the agent does not own a payment card, but that cash is her only available instrument to settle transactions.³⁴ In [Figure 5b](#) we quantify the benefit of holding a payment card, by comparing the cost of managing consumption transaction C with its counterfactual for an agent who is not endowed with a payment card. Notice that R, b, κ and u are held fixed to their estimated values; therefore, any difference in costs we find is attributable to the different policies followed by the agent when she does not have access to a payment card. The results suggest that owning a payment card enables households to save around 70% on the total cost of managing consumption transactions. A household with no access to a payment card would spend 50 euros per year to manage consumption transactions (instead of 15): thus, the value of owning a payment card is around 35 euros per year.³⁵ The savings are mainly driven by the lower opportunity costs of holding cash (as having a payment card reduces the precautionary motive for keeping cash balances high) and by a lower disutility from lost purchases, while the decrease in withdrawal costs plays a negligible role.

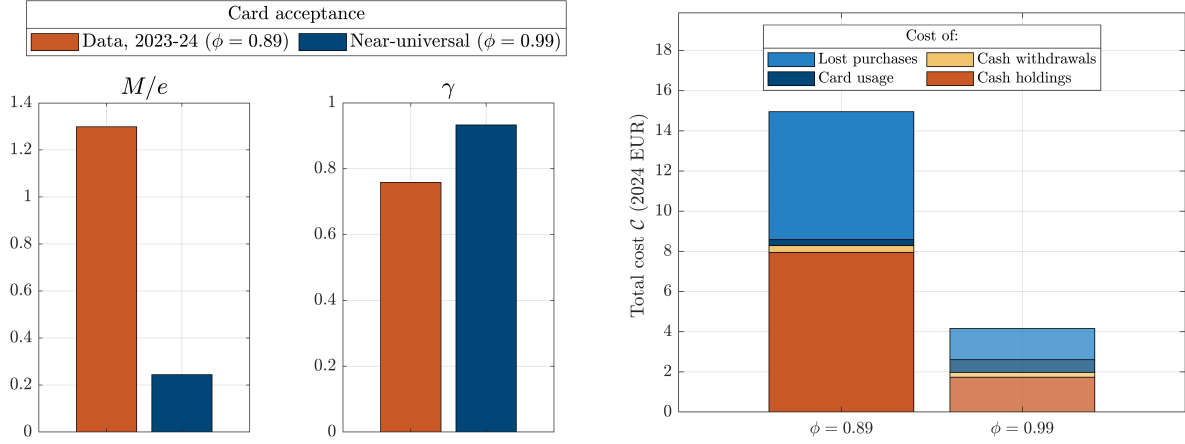
5.2 Near-universal card acceptance

We now analyze a counterfactual scenario in which 99% of all merchants accept payment cards, a scenario we refer to as “near-universal card acceptance”. We start by comparing two key moments that summarize cash management and payment method choices by households. The left panel of [Figure 6](#) shows both the model-implied average cash balance and the card share of expenditure in 2023-24 under the true card acceptance rate $\phi = 0.89$, and when we set $\phi = 0.99$. In the latter scenario the average cash holdings (relative to daily expenditures) decrease by around 110pp, from the current 130% of daily expenditure to about 20%. This happens since with high card acceptance the agent resets her cash balances to a lower level m^* upon withdrawing, and the trigger cash level $\underline{m}$ decreases as well, since the insurance motive

³⁴We implement this by setting $\phi = 0$, i.e., by assuming that no merchants accept payment cards.

³⁵This figure aligns with the cost of a bank account with a payment card in Italy, which was estimated at around 30-40 EUR/year in 2023 ([Banca d'Italia, 2024](#)).

FIGURE 6: Near-universal card acceptance: a counterfactual.

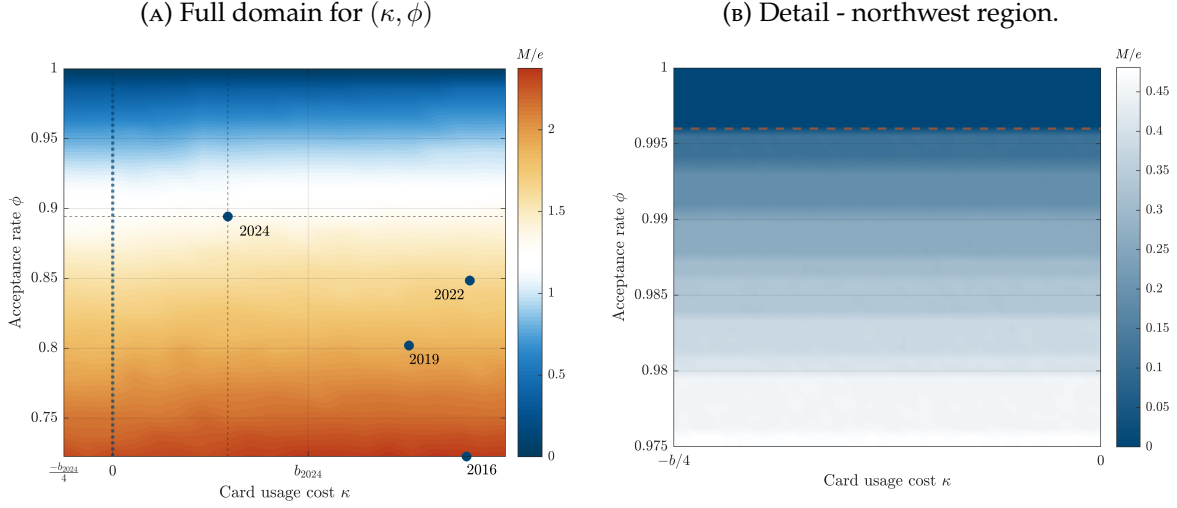


Note: The left panel compares counterfactual moments obtained by solving the estimated model for 2023-24 with $\phi = 0.99$ and their real-world counterparts under the true acceptance rate $\phi = \hat{\phi} = 0.89$. Displayed moments are the average cash balances relative to daily expenditure M/e , and the share of expenditure settled using cards γ . The right panel shows the total cost of managing consumption transactions $\mathcal{C}$ (expressed in 2024 euros), both for the estimated model and for the alternative scenario with $\phi = 0.99$.

(associated with avoiding missed purchase opportunities) is now less relevant. The share of card expenditure by 17pp to about 93%, mostly due to an increase in the share of forced card transactions, as agents more frequently lack sufficient cash to complete their purchases. We find that the share of *unforced* card expenditures falls by around 10pp. This happens because, as acceptance rises, the precautionary motive to use cards when cash is an option (i.e., preserving a buffer-stock of cash balances) is weaker.

We then compute the welfare costs of imperfect acceptance of payment cards for households by comparing the estimated cost of managing consumption transactions $\mathcal{C}$ with its counterfactual value under near-universal card acceptance. The right panel of Figure 6 illustrates the welfare loss that imperfect acceptance generates with respect to the counterfactual world with $\phi = 0.99$: under near-universal acceptance the yearly cost of managing consumption transaction would be about 4 euros, almost 75% smaller than the baseline cost. The difference is mainly due to savings on the opportunity costs of holding cash and to the lower frequency of missed purchases.

FIGURE 7: Conditions for a no-cash policy.



Note: The left panel displays the model-implied moment M/e , i.e., average cash holdings normalized by daily expenditure for goods and services, in a set of economies with all parameters except κ and ϕ set to their estimated levels for 2023-24. We solve the model for a rectangular array of $\{\kappa, \phi\}$, with $\kappa \in [-b_{2024}/4, b]$ and $\phi \in [\hat{\phi}_{2015-16}, 1]$. The dashed blue line separates the right region, in which paying with cards is more expensive than using cash, from the left region, in which cards are a cheaper means of payment. The values of $\{\hat{\kappa}, \hat{\phi}\}$ estimated for 2015-16, 2019 and 2021-22 are overlaid to the plot. The dark blue area with $M/e = 0$ denotes the region in the space of (κ, ϕ) in which no-cash policies are optimal. The right panel zooms into the region where $\kappa \leq 0$ and $\phi \geq 0.975$ (bordered in orange at the northwest of the right panel). The colors are changed to ease visualization of the boundary (dashed orange line) of the set of parameters for which $M/e = 0$.

5.3 A cashless economy?

An interesting question is whether cash can be completely replaced by alternative payment technologies, and under which circumstances (see [Shy \(2023\)](#)). Our model allows us to investigate the conditions under which individuals stop holding cash and use their payment cards to settle all purchases. The cost of managing consumption transactions under a *cashless* policy is $\mathcal{C}_{\text{cashless}} = \lambda(\phi\kappa + (1 - \phi)u)$, as individuals will perform no withdrawals nor will they bear any opportunity costs of holding cash: their only costs are that stemming from card fees when meeting a merchant that accepts cards, and from missed purchase opportunities otherwise. We estimate $\mathcal{C}_{\text{cashless}}$ to be approximately 75 euros per year, around 5 times larger than the cost of managing consumption transactions under the optimal mix of cash and cards, and about 25 euros higher than the cost of following a cash-only policy, quantified in [Section 5.1](#).³⁶ Our model indicates that a complete ban on cash (as discussed by [Rogoff](#)

³⁶Under a cashless policy, the average consumer would miss approximately 80 transactions per year (about 10% of total purchase opportunities) because of imperfect acceptance of cards. Multiplied by the estimated average

(2017) and Alvarez, Argente, et al. (2022)) would increase the cost of managing consumption transactions by 60 euros per year for the average consumer. We next perform the following exercise: using our parameter estimates we construct counterfactuals in which we vary the cost of using cards (κ) and the share of merchants that accept cards (ϕ). We solve the model for all values of ϕ higher than $\hat{\phi}_{2015-16}$, and for a vector of $\kappa \geq b_{2024}/4$. We also include some negative values of κ to take into account the possibility that card payments become cheaper than cash ones (recall that the cost of a cash payment is set to zero in our model): it is worth exploring the solution of the model for negative values of κ as in many surveys, a sizeable fraction of individuals indicate cards as their preferred payment method, as documented by van der Cruysen, Hernandez, and Jonker (2017), among others, and as our data shows too.³⁷ For each $\{\kappa, \phi\}$ -pair, we analyze under which combinations of these two parameters the transactions demand for cash by households becomes zero: we look at the model-implied average cash balances M/e , with $M/e = 0$ identifying a cashless economy. We display the results in Figure 7. The Figure shows that two conditions are necessary for the agent to completely abandon cash: universal or near-universal acceptance (ϕ above 99%), and cheap card payments (small κ), possibly even cheaper than cash ones (negative κ). A very high ϕ is needed for consumers to abandon cash since the estimated value of u , the cost of missing purchases, is quite large. Recall from Proposition 1 that cash is never used to pay for unforced purchases if $\kappa \leq 0$; therefore, the agent follows a no-cash policy whenever card acceptance is universal ($\phi = 1$) and $\kappa \leq 0$.³⁸ Notice that for $\kappa \leq 0$, M/e is insensitive to κ : the agent always pays with her card whenever she can, so M/e only depends on ϕ , i.e., on how often she runs into stores accepting cards. In the Figure, we also plot the values $\{\hat{\kappa}, \hat{\phi}\}$ for 2023-24, as well as those obtained by calibrating the model to match moments for 2015-16, 2019 and 2021-22: the estimates suggest that euro area households are getting closer to a no-cash policy over time, mainly because of increased merchant acceptance. With the currently estimated card usage

cost of missing a transaction (0.97 euros), the associated cost would amount to about 75 euros, accounting for more than 90% of total costs.

³⁷SUCH and SPACE respondents were also asked about their payment method preferences through the question “If you were offered various payment methods in a shop, what would be your preference?” We display the responses for all survey years in Figure A.6. A large fraction of households in the euro area reportedly prefers to pay using cards.

³⁸While this is a sufficient condition for $M/e = 0$, we show in the right panel of Figure 7 that it is not a necessary condition, as $M/e = 0$ for all $\phi > 0.99$.

costs, though, even near-universal acceptance of cards by vendors would not be enough to produce a cashless economy.

6 Concluding remarks

This paper used granular data from payment diaries collected by the ECB to uncover new patterns in the payment choices of euro area households. We empirically show some limitations of the predictions of existing payment choice models. Transaction-level data illustrates that both the amount of cash on hand at the moment of a given purchase, and the size of the incoming purchase, are relevant predictors of decisions between cash and cards. Moreover, we show that a key determinant of card usage is a precautionary motive for holding cash: whenever the incoming purchase is large enough to deplete the consumer's cash balance, cards are more likely to be used. Finally, we document a relationship between card acceptance rates, i.e., the share of merchants willing to accept card payments, and cash management by households: in areas with lower card acceptance, individuals hold a precautionary buffer stock of cash, visiting ATMs when their wallets are still half-full.

We rationalize the above patterns through an inventory-theoretical model with uncertain lumpy expenditures of random size and a choice between cash and cards. The agent can decide whether to settle her purchases using a payment card (whenever she meets a merchant who accepts card payments), or with cash. Using cards involves the payment of a fixed cost, but keeps cash balances at a higher level. Whenever the cost of using cards is smaller than that of visiting ATMs, the agent sometimes optimally uses her card to pay even though she has sufficient cash on hand: in particular, whenever the cost of using cards is smaller than the increase in the value of the problem resulting from a cash payment, cards are employed. The optimal policy shares features of [Whitesell \(1989\)](#) (transaction-threshold type payment choices) and [Alvarez and Lippi \(2017\)](#) (choices depend on current cash holdings), rationalizing the micro-level evidence on payments of [Section 2](#) in a unified framework.

We calibrate the model to replicate features of observed payments behavior in the euro area. The estimated cost of using cards is smaller than the cost of a cash withdrawal. We use the model to perform several counterfactual analyses, quantifying the costs of managing

consumption transactions, the benefits of card ownership, the welfare costs of imperfect acceptance, and providing conditions under which a cashless economy might emerge.

There are three main caveats in the interpretation of our results, which stem from the simplicity of the model and that could be addressed in a more complete framework of analysis. First, in the model there are no exogenous inflows of cash: in order to obtain cash, the agent has to pay a fixed cost and she always resets cash balances to the optimal level. The data however show that households oftentimes receive part of their income as cash or obtain cash from friends, relatives and colleagues: including exogenous cash inflows in the model (adopting a framework a' la [Miller and Orr \(1966\)](#)) would incorporate this feature, which might be important in determining the resilience of cash. Second, our model does not allow the relative cost κ of using cards to depend on the transaction size s . There are, however, reasons why this assumption may be too restrictive. For instance, credit-card reward programs reimburse consumers for a portion of their card-based expenditures: in such cases, the effective cost of using cards decreases as s rises. A possible extension of our framework could allow for a more flexible card-usage cost, such as $\kappa(s) = \kappa_0 + \kappa_s s$, which could yield more nuanced implications for the substitutability between cash and cards. This extension may be particularly relevant when calibrating the model for an economy where reward programs are widespread, such as the United States. Third, our analysis focussed on the consumers' side of the problem, taking the share of stores that accept cards as given. This affects our computation of the welfare costs of imperfect card acceptance, as we neglect the potentially negative impact that full acceptance of payment cards could have on those merchants who are only willing to receive cash payments as of now.³⁹ Future work might entail modeling card acceptance choices as a competition game between sellers (as in [Masters and Rodríguez-Reyes \(2005\)](#)) potentially aiming to attract buyers that sort into shops based on their payment methods acceptance policies (see [Moracci and Sorbera \(2022\)](#) for an example), evaluating which margins determine card acceptance rates, payment choices and cash management in equilibrium.

³⁹It has been shown that card networks design fee structures which impose a greater share of transaction costs on merchants (see [Rochet and Tirole \(2014\)](#), [Huynh, Nicholls, and Shcherbakov \(2022\)](#)), discouraging some stores from accepting card payments.

References

- Alvarez, Fernando, David Argente, et al. (2022). "Cash: A Blessing or a curse?" In: *Journal of Monetary Economics* 125, pp. 85–128.
- Alvarez, Fernando and Francesco Lippi (2009). "Financial Innovation and the Transactions Demand for Cash". In: *Econometrica* 77.2, pp. 363–402.
- (2013). "The Demand of Liquid Assets with Uncertain Lumpy Expenditures". In: *Journal of Monetary Economics* 60.7, pp. 753–770.
- (2017). "Cash Burns: An Inventory Model with a Cash-Credit Choice". In: *Journal of Monetary Economics* 90, pp. 99–112.
- Arango, Carlos, Kim P. Huynh, Ben Fung, et al. (2012). "How Do You Pay? The Role of Incentives at the Point-of-Sale". In: *Bank of Canada Review* (Autumn), pp. 31–40.
- Arango, Carlos, Kim P. Huynh, and Leonard Sabetti (2015). "Consumer Payment Choice: Merchant Card Acceptance versus Pricing Incentives". In: *Journal of Banking and Finance* 55. February 2015, pp. 130–141.
- Arango, Carlos A., Yassine Bouhdaoui, et al. (2014). "Cash Management and Payment Choices: A Simulation Model with International Comparisons". In: *SSRN Electronic Journal*.
- Arango-Arango, Carlos A. et al. (2018). "Cash Remains Top-of-Wallet! International Evidence from Payment Diaries". In: *Economic Modelling* 69. October 2017, pp. 38–48.
- Bagnall, John et al. (2016). "Consumer Cash Usage: A Cross-Country Comparison with Payment Diary Survey Data*". In: *International Journal of Central Banking* 12.4, pp. 1–61.
- Banca d'Italia (Dec. 2024). *Indagine sul costo dei conti correnti nel 2023*. Tech. rep. n. 14/2024. Rilevazione condotta su 11.985 conti correnti bancari, 1.174 conti online e 1.000 conti postali. Roma, Italia: Banca d'Italia.
- Bar-Ilan, Avner, David Perry, and Wolfgang Stadje (Mar. 2004). "A Generalized Impulse Control Model of Cash Management". In: *Journal of Economic Dynamics and Control* 28.6, pp. 1013–1033.
- Baumol, William J. (1952). "The Transactions Demand for Cash: An Inventory Theoretic Approach". In: *The Quarterly Journal of Economics* 66.4, p. 545.

- Beckmann, Martin (1961). "An Inventory Model for Arbitrary Interval and Quantity Distributions of Demand". In: *Management Science* 8.1, pp. 35–57.
- Bouhdaoui, Yassine and David Bounie (2012). "Modeling the Share of Cash Payments in the Economy: An Application to France". In: *International Journal of Central Banking* 8.4, pp. 175–195.
- Briglevics, Tamas and Scott Schuh (2021). "This Is What's in Your Wallet... And How You Use It". In: *SSRN Electronic Journal* 14.
- Ching, Andrew T and Fumiko Hayashi (2010). "Payment Card Rewards Programs and Consumer Payment Choice". In: *Journal of Banking and Finance* 34, pp. 1773–1787.
- Eppen, Gary D. and Eugene F. Fama (1969). "Cash Balance and Simple Dynamic Portfolio Problems with Proportional Costs". In: *International Economic Review* 10.2, p. 119.
- Eschelbach, Martina and Tobias Schmidt (2013). "Precautionary Motives in Short-Term Cash Management: Evidence from German Pos Transactions". In: *SSRN Electronic Journal*.
- Freeman, Scott and Finn E. Kydland (2000). "Monetary Aggregates and Output". In: *American Economic Review* 90.5, pp. 1125–1135.
- Huynh, Kim, Gradon Nicholls, and Oleksandr Shcherbakov (2022). "Equilibrium in Two-Sided Markets for Payments: Consumer Awareness and the Welfare Cost of the Interchange Fee". In.
- Huynh, Kim P., Philipp Schmidt-Dengler, and Helmut Stix (2014). "The Role of Card Acceptance in the Transaction Demand for Money". In: *Oesterreichische Nationalbank Working Papers* 49.196, pp. 1–40.
- Klee, Elizabeth (2008). "How People Pay: Evidence from Grocery Store Data". In: *Journal of Monetary Economics* 55.3, pp. 526–541.
- Knümann, Fabio, Malte Krüger, and Franz Seitz (2024). *Costs of Cash and Card Payments from a Consumer Perspective*. Tech. rep. Module 3, Costs and Benefits of Cash and Cashless Payment Instruments. English edition published in 2025. Frankfurt am Main, Germany: Deutsche Bundesbank.
- Lucas, Robert and Nancy Stokey (1987). "Money and Interest in a Cash-in-Advance Economy". In: *Econometrica* 55.3, pp. 491–513.

- Masters, Adrian and Luis Raúl Rodríguez-Reyes (2005). "Endogenous Credit-Card Acceptance in a Model of Precautionary Demand for Money". In: *Oxford Economic Papers* 57.1, pp. 157–168.
- Miller, Merton H. and Daniel Orr (1966). "A Model of the Demand for Money by Firms". In: *The Quarterly Journal of Economics* 80.3, p. 413.
- Moracci, Elia and Silvio Sorbera (2022). "Payment Choices and Cash Management with Endogenous Merchant Acceptance".
- Nocciola, Luca and Alejandro Zamora-Peréz (2024). *Transactional demand for central bank digital currency*. Tech. rep. 2926.
- Perera, Sandun C. and Suresh P. Sethi (2023). "A Survey of Stochastic Inventory Models with Fixed Costs: Optimality of (s, S) and (s, S)-Type Policies—Continuous-time Case". In: *Production and Operations Management* 32.1, pp. 154–169.
- Prescott, Edward C. (1987). "A Multiple Means-of-Payment Model". In: *New approaches to monetary economics*, pp. 42–51.
- Rochet, Jean Charles and Jean Tirole (2014). "Platform Competition in Two-Sided Markets". In: *Competition Policy International* 10.2, pp. 180–218.
- Rogoff, Kenneth S (2017). *The Curse of Cash*.
- Schuh, Scott and Joanna Stavins (2016). "HOW DO SPEED AND SECURITY INFLUENCE CONSUMERS' PAYMENT BEHAVIOR?" In: *Contemporary Economic Policy* 34.4, pp. 595–613.
- Shy, Oz (2023). "Cash is alive: How economists explain holding and use of cash". In: *Journal of Economic Literature* 61.4, pp. 1465–1520.
- Stix, Helmut (2004). "How Do Debit Cards Affect Cash Demand? Survey Data Evidence". In: *Empirica* 31.2, pp. 93–115.
- Telyukova, Irina A. (2013). "Household Need for Liquidity and the Credit Card Debt Puzzle". In: *The Review of Economic Studies* 80.3 (284), pp. 1148–1177.
- Tobin, James (1956). "The Interest-Elasticity of Transactions Demand For Cash". In: *The Review of Economics and Statistics*.
- Trütsch, Tobias (2020). "The impact of contactless payment on cash usage at an early stage of diffusion". In: *Swiss Journal of Economics and Statistics* 156.5, pp. 1–35.

- van der Cruijssen, Carin, Lola Hernandez, and Nicole Jonker (2017). "In Love with the Debit Card but Still Married to Cash". In: *Applied Economics* 49.30, pp. 2989–3004.
- Wakamori, Naoki and Angelika Welte (2017). "Why Do Shoppers Use Cash? Evidence from Shopping Diary Data". In: *Journal of Money, Credit and Banking* 49.1, pp. 115–169.
- Wang, Zhu and Alexander L. Wolman (2016). "Payment Choice and Currency Use: Insights from Two Billion Retail Transactions". In: *Journal of Monetary Economics* 84, pp. 94–115.
- Whitesell, William C. (1989). "The Demand for Currency versus Debitable Accounts: Note". In: *Journal of Money, Credit and Banking* 21.2, p. 246.

Online Appendix to

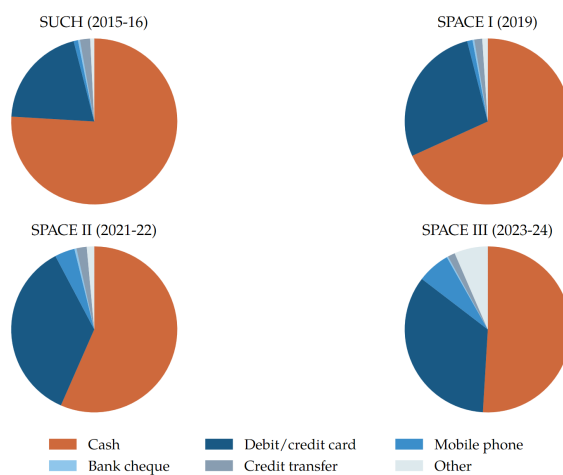
Cash or card? A structural model of payment choices

by **Francesco Lippi** (LUISS and EIEF) and **Elia Moracci** (Bank of Italy)

A Empirical appendix

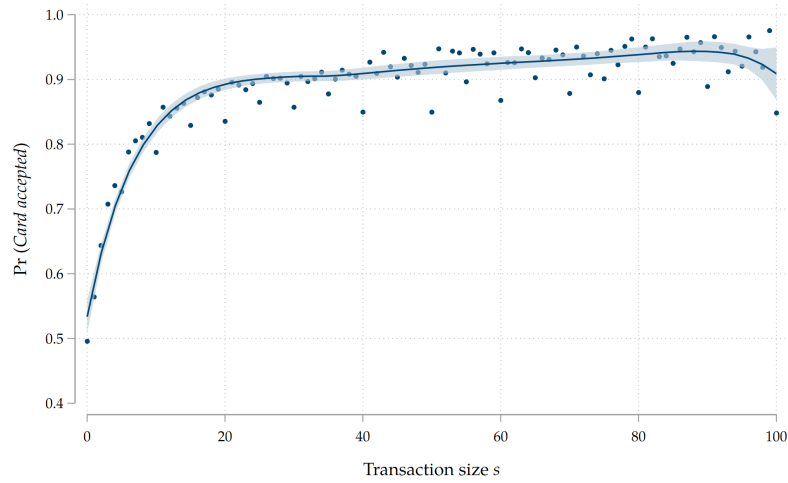
A.1 ECB payment diaries - additional figures

FIGURE A.1: Payment methods usage in different waves.



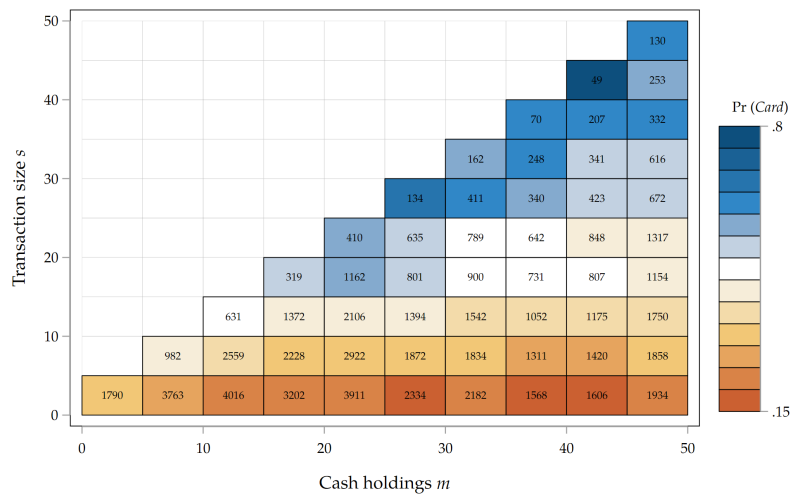
Note: This plot shows the share of transactions settled with each payment method across the three waves. Credit and debit cards are bundled together for comparability, as in the first wave of SPACE it was not asked if the payment card used was a debit or a credit card.

FIGURE A.2: Card acceptance for different transaction sizes s



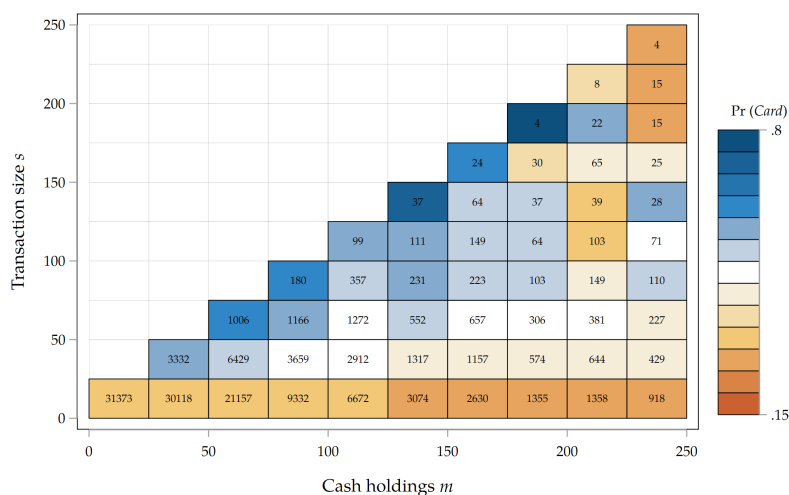
Note: Transaction sizes are rounded to the closest integer. For each binned transaction size, we plot the share of payments for which card payments were accepted at the point of sale. A third degree polynomial fit is overlaid to the plot.

FIGURE A.3: Share of cash payments for different m and s . Zooming in on smaller transactions/cash holdings levels.



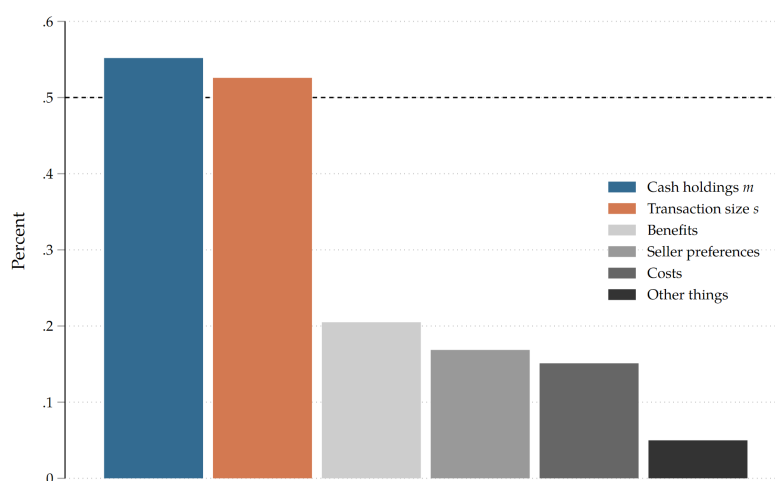
Note: The Figure displays the share of payments settled using cards for bins defined in terms of cash holdings at payment (m) and transaction size faced (s). Numbers denote the number of observations falling in each bin. We focus on transactions where m and s are smaller or equal than 50 euros.

FIGURE A.4: Share of cash payments for different m and s . Including larger transactions and cash holdings.



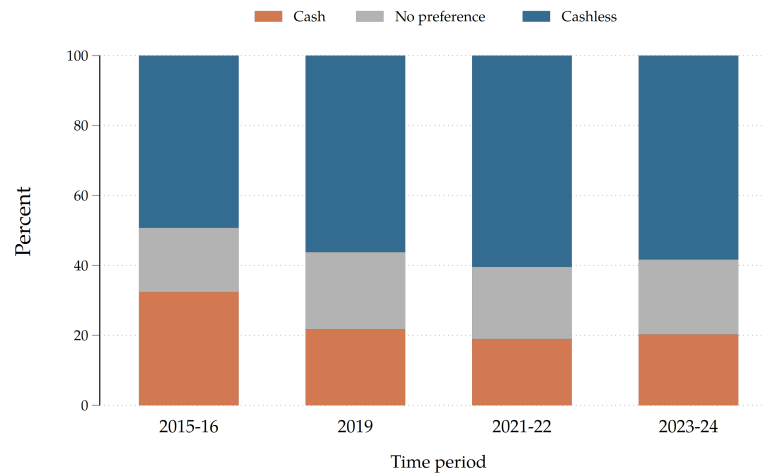
Note: The Figure displays the share of payments settled using cards for bins defined in terms of cash holdings at payment (m) and transaction size faced (s). Numbers denote the number of observations falling in each bin. We focus on transactions where m and s are smaller or equal than 250 euros.

FIGURE A.5: Reported determinants of payment method decisions.



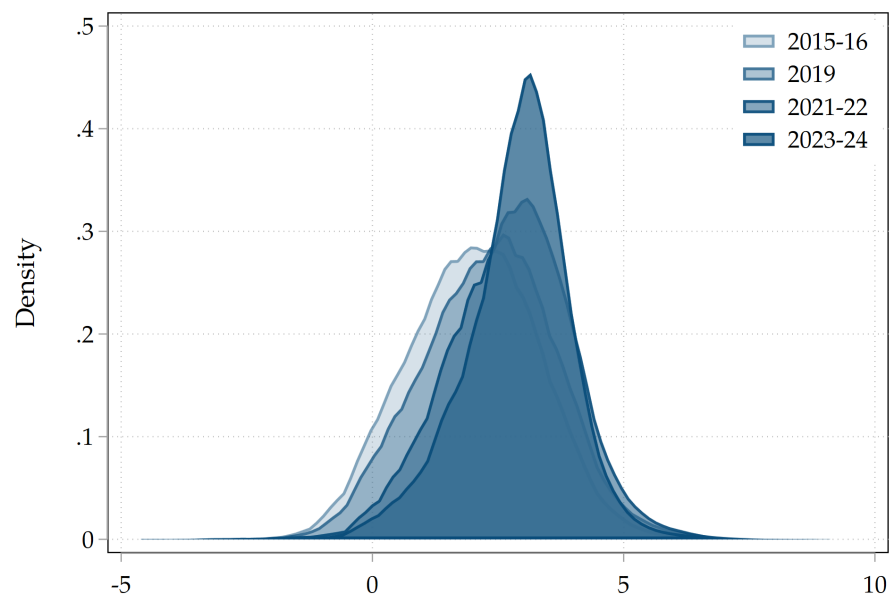
Note: This graph displays the shares of households reporting that a certain factor influences their payment decision. The question respondents answered was: "Which of the following influences your decision to pay with cash or card or other non-cash payment methods?". Multiple responses are possible. Source: ECB SUCH (2016) Data.

FIGURE A.6: Payment preferences over the years.



Note: This Figure shows the yearly distribution of answers to the question “If you were offered various payment methods in a shop, what would be your preference?”.

FIGURE A.7: Payment size distribution (EUR, logs)



A.2 Payment choice sets in SUCH and SPACE data

We now describe how to construct transaction-level *payment choice sets* using the SUCH and SPACE payment diary data, distinguishing between *forced cash*, *forced card*, and *unforced* payments.

The ECB payment diaries do not record cash balances at the exact time of each transaction. Let m_{it} denote the amount of cash held by individual i immediately before her t th transaction, and let s_{it} denote the corresponding transaction size. Cash holdings at payment are reconstructed starting from reported cash balances at the beginning of the diary day and updating them using information on cash withdrawals, cash receipts, and cash payments recorded during the day.

Two data limitations require additional discussion. First, for almost all respondents (with the exception of few participants in the SUCH survey which were explicitly asked) the exact timing of cash replenishments is not observed. In these cases, the timing of withdrawals can be inferred, to some extent, whenever a cash payment would otherwise violate the feasibility condition $s_{it} \leq m_{it}$. Purchases for which we could not uniquely reconstruct cash holdings at the time of the transaction were flagged, and excluded from all analyses at the transaction level. Second, we don't have information on the presence and timing of cash deposits during the diary day. We therefore compute predicted end-of-day cash balances assuming no deposits and compare them with reported end-of-day balances; observations implying unreported deposits or negative cash balances at any point during the day were dropped.

We excluded online payments and restricted the sample to respondents with access to at least one payment card. After applying these restrictions, the final sample includes a total of 338,773 transactions (carried out with either cash or cards) performed by 184,565 individuals.

For each purchase for which we are able to pin down cash balances at the time of the transaction, we determine the payment choice set (which payment methods were feasible) as follows. Cash is feasible if the transaction size does not exceed available cash holdings ($s_{it} \leq m_{it}$) and if the merchant accepts cash.⁴⁰ Card payments are feasible if the merchant

⁴⁰In the SUCH survey, respondents were not asked if cash was accepted at the store. We therefore assume that cash was always accepted by merchants in 2015-16, which is consistent with the near-universal levels of cash

accepts at least one card-based payment instrument and the respondent owns a debit card, a credit card, or both. Since the data do not identify which specific types of cashless payment methods are accepted at each transaction, we assume that debit and credit cards are accepted whenever respondents report that cashless payments were accepted at the store, while we consider the access to other cashless instruments (e.g., cheques or credit transfers) as not sufficient to guarantee the feasibility of a card payment.

A transaction is classified as *unforced* if both cash and card payments are feasible. Otherwise, the transaction is classified as either *forced cash* or *forced card* depending on which payment method is available. For some transactions, we cannot pin down the payment choice set even if we observe m_{it} with certainty: this happens when (i) the size of the transaction s_{it} is missing, or (ii) the respondent did not state whether the alternative payment method (with respect to the one they used) was accepted.

A.3 Payment choices: regression analysis

In this Subsection, we test if the descriptive findings of [Section 2](#) are robust to a more sophisticated analysis. In principle, it could be that patterns emerging in [Figure 2](#) are entirely due to selection: for instance, it might be that households who generally prefer to use cards have on average lower cash holdings, or that they buy more valuable goods. As our data contains multiple transactions per individual, we can rule out these potential issues by relying on fixed effects models. In particular, we want to assess whether i) higher s increases the probability of a cashless payment, ii) higher m decreases the probability of a cashless payment; iii) the effects of a rise in s is stronger when m is smaller and viceversa, as m' acceptance observed in the earlier SPACE waves.

TABLE 3: Regression evidence on the joint importance of m and s .

Unit: EUR 100	Dependent variable: $PayCard_{it}$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cash holdings m	-0.11*** (0.0014)	-0.098*** (0.0014)	-0.037*** (0.0018)	-0.058*** (0.0022)	-0.052*** (0.0017)	-0.050*** (0.0017)	-0.0038 (0.0022)	-0.020*** (0.0025)
Payment size s	0.61*** (0.0040)	0.47*** (0.0042)	0.31*** (0.0077)	0.41*** (0.0076)	0.79*** (0.0060)	0.62*** (0.0060)	0.54*** (0.012)	0.68*** (0.012)
Cash holdings $m \times$ Payment size s					-0.20*** (0.0047)	-0.16*** (0.0044)	-0.15*** (0.0060)	-0.18*** (0.0063)
Observations	251427	229694	137183	150839	251427	229694	137183	150839
Unforced			✓	✓			✓	✓
Controls		✓	✓			✓	✓	
Random effects				✓				✓
Robust SEs	✓	✓	✓	✓	✓	✓	✓	✓

Note: Columns (2-3) and (6-7) include controls such as country, survey year, sex, age group, education and income of respondents, type of store. Columns (3-4) and (7-8) only take into account unforced transactions, i.e., transactions where both payments methods were available for the respondent. Columns (4) and (8) include individual-level random effects. Heteroskedasticity-robust standard errors are reported.

matters for choices.⁴¹ We estimate linear probability models of the form

$$PayCard_{it} = \beta_0 + \beta_s s_{it} + \beta_m m_{it} + \beta_{sm} (s_{it} \times m_{it}) + \lambda' \mathbf{X}_{it} + \zeta_i + \varepsilon_{it}, \quad (13)$$

where $PayCard_{it} = 1$ ($= 0$) if individual i settles her t -th transaction using cashless methods (cash), $\mathbf{X}_{it}$ denote transaction-specific characteristics and ζ_i denote individual-level fixed effects.

The results are displayed in Table 3. Models (1) to (4) are intended to test whether card usage probabilities are increasing in the transaction size s and decreasing in the amount of cash holdings m . Our preferred specification (column (4), where we focus on unforced transactions and we include individual-specific random effects) reveals that a EUR 10 increase

⁴¹To test if choices are affected by m' , we can estimate whether the effect of s is different according to the level of m . If s only affects choices through perceived cost/benefits of using cash instead of cards (for instance, because there are proportional fees, or because agents dislike paying small sums using their card as they perceive that merchants prefer them to pay cash), the effect of s should be independent of m , as long as $s \leq m$. At the same time, if agents care about the level of m because they want to get rid of cash they have, the effect of m shouldn't depend on s , as long as $s \leq m$. The presence of a significant interaction term points towards the existence of a more complex relationship between s , m and choices, which would be consistent with the story suggested by the previous two Figures. If households don't want to run out of cash, higher s increases the probability of cashless payments, but the effect of s is smaller as m grows. At the same time, higher m decreases the probability of a cashless payment, and it decreases it more when s is larger (as one abandons the region when $m - s \simeq 0$).

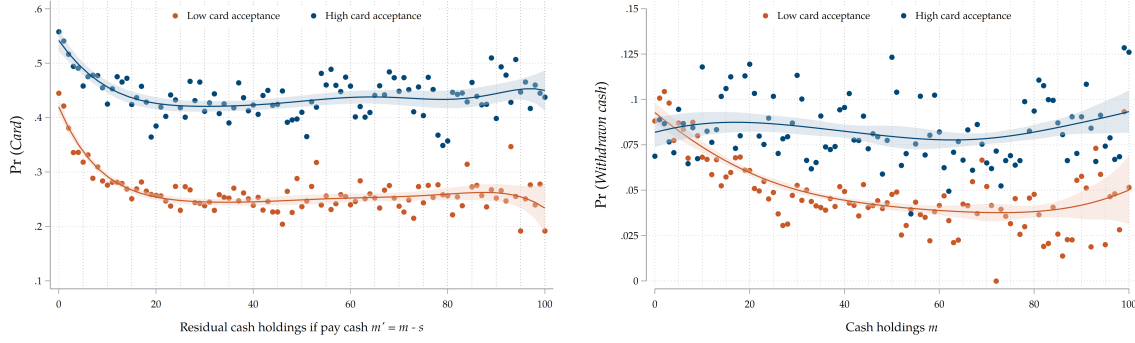
in cash holdings m is associated with 0.6pp decrease in the probability of paying by card. The opposite is true for the transaction size: a 10 EUR increase in s is associated with a 4.1pp rise in the probability of card usage. Models (5) to (8) are instead meant to test whether choices are really driven by $m' = m - s$. In order to do so, we can estimate whether the effect of m changes depending on the level of s . In particular, if agents really want to avoid running out of cash, thereby increasingly relying on cards as $m' \rightarrow 0$, we should find a negative coefficient on the interaction between cash holdings and payment size⁴². The estimated coefficient is indeed negative and statistically significant, corroborating our descriptive analysis. In particular, we find that the probability of a card payment decreases by 3.2pp for a EUR 10 increase when the payment size s is close to zero, but the marginal impact of cash holdings becomes stronger as the transaction size rises: for instance, for a EUR 50 payment, having 10 more euros on hand is associated with an extra probability of paying with cards of 4pp. Similarly, notice that the effect of payment size decreases as m rises, and even turns negative for sufficiently high cash holdings. Notice that this is consistent with optimal cash holdings being finite: if agents optimally want to hold a quantity of cash m^* , for $m > m^*$ they will increasingly use cash as s increases, as when cash holdings are very high, large payments can be exploited to bring cash holdings closer to their optimal value.

A.4 The role of imperfect card acceptance

The evidence of [Section 2](#) shows that households prefer to avoid having too little cash and therefore use cards when doing otherwise would result in a near depletion of their cash balances. This suggests the existence of a precautionary motive to hold a buffer stock of cash. A possible driver of such precautionary motive is avoiding situations in which cash balances are too low to carry out a transaction, especially if there is a significant risk that cards are not accepted at the point of sale. There are two ways to avoid little cash on hand: agents can either use cards when s is very close to m or visit ATMs way before they reach $m = 0$. Our data enables us to understand if the households' precautionary motive for holding cash is driven (at least to some extent) by imperfect card acceptance. We do this

⁴²If households do not want to run out of cash, higher s increases the probability of card payments, but the effect of s is smaller as m grows. At the same time, higher m decreases the probability of a card payment, and it decreases it more when s is larger (as one abandons the region when $m - s \simeq 0$).

FIGURE A.8: Card acceptance, payment choices and cash management.



Note: The left panel displays the shares of households paying using cards for bins defined in terms of cash holdings remaining in case agents settle the payment using cash (*implied residual cash holdings* $m' = m - s$), for two groups of individual respectively exposed to a card acceptance rate lower (higher) than the median one. Only unconstrained transactions are considered, and transactions with $m = s$ are omitted. The right panel displays the probability of withdrawing cash as a function of m , i.e., the current level of cash balances. Two nonparametric fits with 95% confidence intervals are overlaid to each plot.

Data from ECB payment diaries: SUCH (2016) and SPACE (2019, 2021-22 and 2023-24).

by splitting the sample in two groups of households according to the card acceptance rate they are exposed to.⁴³ Figure A.8 shows how the payment choices and cash management of these two groups of households differ in the ways we hypothesized above. In the left panel, we display a split-sample version of the right panel of Figure 2. The Figure shows that the probability of using cards rises as $s \rightarrow m$ especially for households exposed to low acceptance rates; in other words, imperfect acceptance is at least partially driving the rise in the probability of card usage as $s \rightarrow m$ we described in Section 2. In the right panel of Figure A.8, we display the probability of performing a cash withdrawal before the next purchase as a function of current cash holdings m , for the two groups of households: the graph shows that for households exposed to lower card acceptance rates the probability of withdrawing cash spikes up when m is low, while this does not apply to households exposed to high card acceptance. Taken together, the two plots suggest that households want to hold low amounts of cash when card acceptance is far from being universal, since being matched with a non-accepting vendor in those circumstances would lead to either i) the impossibility to carry out the desired transaction, or ii) the need to rush to an ATM before going back to the store and completing the purchase.

⁴³Expected acceptance rates are predicted probabilities derived from a logistic regression with demographic and geographical controls. Details are available upon request.

B Theoretical appendix

B.1 Proof of Lemma 1

If $p(m, s) = 0$, it means that $v(m - s) \leq v(m) + \kappa$. But then $v(m - s') \leq v(m - s) \leq v(m) + \kappa$ too if v is weakly decreasing. The second result follows from the same logic. ■

B.2 Proof of Lemma 2

Consider payments of size s such that $m - s \leq \underline{m}$. Agents are indifferent between using cash and cards if $m = \tilde{m}$, where $\tilde{m}$ solves $v(\tilde{m}) = v^* + b - \kappa$ and it is independent of s . As $v(m)$ is decreasing, $p(m, s) = 1$ for $m > \tilde{m}$ and $p(m, s) = 0$ otherwise. A solution to $v(\tilde{m}) = v^* + b - \kappa$ exists if and only if $0 < \kappa < b$, as $v(m) \in [v^*, v^* + b]$. Since i) v is weakly decreasing and continuous, ii) $v(\underline{m}) = v^* + b$, and iii) $v(m^*) = v^*$, by the intermediate value theorem a solution $\tilde{m} \in (\underline{m}, m^*)$ exists, as desired. ■

B.3 Proof of Proposition 1

We now prove each point of Proposition 1.

1. Recall that the agent uses cards whenever $v(m - s) \geq v(m) + \kappa$. When $\kappa \leq 0$, this is always the case as v is weakly decreasing. This proves that for $\kappa \leq 0$, $p(m, s) = 1$ for all (m, s) , as desired. ■
2. (a) From Lemma 2, we know that when $m \in [\underline{m}, \tilde{m}]$, $p(m, s) = 0$ whenever $m - \underline{m} \leq s \leq m$. By Lemma 1, then $p(m, s) = 0$ also for any $0 \leq s < m - \underline{m}$.
(b) From Lemma 2 we know that for $m - s \leq \underline{m}$, we have $p(m, s) = 1$ whenever $m \in [\tilde{m}, m^*]$. By Lemma 1, this implies that there exists $\underline{s}(m) \leq m - \underline{m}$ such that $p(m, s) = 1$ for all $s \geq \underline{s}(m)$, as desired. In particular, we have that $\underline{s}(m) < m - \underline{m}$ whenever

$$v(m - s) > v(m) + \kappa,$$

for some $m > \tilde{m}$ and $s < m - \underline{m}$. Then, for any $m > \tilde{m}$, $\underline{s}(m)$ is the solution to

$$v(m - \underline{s}(m)) = v(m) + \kappa,$$

which can be rewritten as

$$\underline{s}(m) = m - v^{-1}(v(m) + \kappa),$$

where $v^{-1}(v(m) + \kappa)$ is well-defined (it is a singleton) as v is strictly decreasing (hence invertible) on $[\underline{m}, m^*]$. ■

3. First, let's prove that when $\kappa > b$, $p(m, s) = 0$ for all m and for all $m - \underline{m} \leq s \leq m$, i.e., for all s such that $m - s \leq \underline{m}$. If $\kappa > b$, $v(m) > v^* + b - \kappa > v^*$ for any m , hence $p(m, s) = 0$ for all m and all $m - \underline{m} \leq s \leq m$. By [Lemma 1](#), it has to be that $p(m, s) = 0$ even for $0 \leq s < m - \underline{m}$. Therefore, $p(m, s) = 0$ for all m and for any $s \leq m$, as desired. ■.

B.4 Additional results

We state and prove two additional results for the case $0 < \kappa < b$ that are not reported in the main text.

1. $\lim_{m \rightarrow \tilde{m}_+} \underline{s}(m) = \tilde{m} - \underline{m}$.

Proof. $\lim_{m \rightarrow \tilde{m}_+} \underline{s}(m) = \tilde{m} - \lim_{m \rightarrow \tilde{m}_+} v^{-1}(v(m) + \kappa) = \tilde{m} - v^{-1}(v^* + b) = \tilde{m} - \underline{m}$. ■

2. $\lim_{m \rightarrow m^*} \underline{s}'(m) = 1$

Proof. If we differentiate [\(7\)](#) we obtain

$$\underline{s}'(m) = 1 - (v^{-1})'(v(m) + \kappa) v'(m).$$

Using the inverse function theorem, this yields

$$\underline{s}'(m) = 1 - \frac{v'(m)}{v'(v^{-1}(v(m) + \kappa))}.$$

Let $\hat{m}(m)$ be the solution to $v(\hat{m}(m)) = v(m) + \kappa$. This leaves us with

$$\underline{s}'(m) = 1 - \frac{v'(m)}{v'(\hat{m}(m))}.$$

If v is strictly convex for $m > \tilde{m}$, v' is decreasing in absolute value. Given that $\hat{m}(m) < m$, we have that $\underline{s}'(m) > 0$. Since v is strictly decreasing, $\underline{s}'(m) < 1$. Moreover, since $\lim_{m \rightarrow m^*} v'(m) = 0$, we have that $\lim_{m \rightarrow m^*} \underline{s}'(m) = 1$, as desired. ■

B.5 Derivation of Equation (8)

We show how to derive (8), the functional equation whose solution is the stationary distribution of cash holdings $h(m)$. We have that

$$\begin{aligned} h(m, t + \Delta) = & (1 - \lambda\Delta)h(m, t) + \\ & + \lambda\Delta h(m, t) (1 - F(m)) + \\ & + \lambda\Delta h(m, t) \int_0^m f(s) \phi p(m, s) ds + \\ & + \lambda\Delta \int_m^{m^*} h(m', t) f(m' - m) (1 - \phi p(m', m' - m)) dm' \end{aligned}$$

Removing time indices and rearranging, we obtain

$$h(m, t) \lambda \Delta \left(F(m) - \int_0^m f(s) \phi p(m, s) ds \right) = \lambda \Delta \int_m^{m^*} h(m', t) f(m' - m) (1 - \phi p(m', m' - m)) dm'$$

which yields the desired result.

B.6 Derivation of Equation (9)

We show how to derive (9), the functional equation whose solution is the function $\mathcal{T}(m)$ which yields the average time to the next withdrawal as a function of current cash holdings m . We start from a discrete-time version of the equation and then take its continuous time

limit. We have

$$\begin{aligned}
\mathcal{T}(m) &= (1 - \lambda\Delta) (\Delta + \mathcal{T}(m)) + \\
&+ \lambda\Delta (1 - F(m)) (\Delta + \mathcal{T}(m)) + \\
&+ \lambda\Delta \int_0^{m-\underline{m}} f(s) (1 - \phi p(m, s)) [\Delta + \mathcal{T}(m - s)] ds + \\
&+ \lambda\Delta \int_{m-\underline{m}}^m f(s) (1 - \phi p(m, s)) ds \cdot 0 + \\
&+ \lambda\Delta \int_0^m f(s) \phi p(m, s) [\Delta + \mathcal{T}(m)] ds.
\end{aligned}$$

Rearranging, we obtain

$$\begin{aligned}
\mathcal{T}(m)\lambda\Delta \left(F(m) - \int_0^m f(s)\phi p(m, s)ds \right) &= \Delta - \lambda\Delta^2 + \\
&+ \lambda\Delta^2 (1 - F(m)) + \\
&+ \lambda\Delta^2 \int_0^{m-\underline{m}} f(s) (1 - \phi p(m, s)) ds + \\
&+ \lambda\Delta \int_0^{m-\underline{m}} f(s) (1 - \phi p(m, s)) \mathcal{T}(m - s) ds + \\
&+ \lambda\Delta^2 \int_0^m f(s)\phi p(m, s)ds.
\end{aligned}$$

Dividing everything by Δ we get

$$\begin{aligned}
\mathcal{T}(m)\lambda \left(F(m) - \int_0^m f(s)\phi p(m, s)ds \right) &= 1 - \lambda\Delta + \\
&+ \lambda\Delta (1 - F(m)) + \\
&+ \lambda\Delta \int_0^{m-\underline{m}} f(s) (1 - \phi p(m, s)) ds + \\
&+ \lambda \int_0^{m-\underline{m}} f(s) (1 - \phi p(m, s)) \mathcal{T}(m - s) ds + \\
&+ \lambda\Delta \int_0^m f(s)\phi p(m, s) [\Delta + \mathcal{T}(m)] ds,
\end{aligned}$$

and taking limits for $\Delta \rightarrow 0$ we finally obtain

$$\mathcal{T}(m) = \frac{1 + \lambda \int_0^{m-\underline{m}} f(s) (1 - \phi p(m, s)) \mathcal{T}(m - s) ds}{\lambda [F(m) - \int_0^m f(s)\phi p(m, s)ds]},$$

or, in terms of implied cash remaining $m' = m - s$,

$$\mathcal{T}(m) = \frac{1 + \lambda \int_{\underline{m}}^m f(m - m') (1 - \phi p(m, m - m')) \mathcal{T}(m') dm'}{\lambda [F(m) - \int_0^m f(m - m') \phi p(m, m') dm']}.$$

B.7 Model-implied moments: additional details

Here, we provide additional details on the computation of model-implied moments, complementing [Section 3.2](#).

Computation of $\underline{M}$ and W . We now illustrate how to compute the average cash on hand when a withdrawal takes place, denoted by $\underline{M}$, as well as the average size of withdrawals W . To do that, we need to compute the stationary distribution of cash balances the moment *before* a payment that triggers a withdrawal takes place. Such a distribution is denoted by $h_w(m)$ and given by

$$h_w(m) = \frac{h(m) \left(\int_{m-\underline{m}}^m f(s) (1 - \phi p(m, s)) ds \right)}{\int_{\underline{m}}^{m^*} h(m) \left(\int_{m-\underline{m}}^m f(s) (1 - \phi p(m, s)) ds \right) dm + h(m^*) \left(\int_{m^*-\underline{m}}^{m^*} f(s) (1 - \phi p(m^*, s)) ds \right)},$$

where the numerator gives the flows into withdrawing coming from cash holdings m and the denominator represents aggregate flows into withdrawing. The boundary condition $\int_{\underline{m}}^{m^*} h_w(m) dm + h_w(m^*) = 1$ helps us pinning down the mass point $h_w(m^*)$. We can then use this probability distribution to compute $\underline{M}$. An extra step is involved: for any m , we should compute the expected value of s given that a withdrawal took place after a payment before which the individual had m on hand, and subtract it from m to obtain the amount of cash holdings $m - s$ right before the withdrawal took place (i.e., after the payment was settled using cash). We obtain

$$\underline{M} = \int_{\underline{m}}^{m^*} h_w(m) \left[\frac{\int_{m-\underline{m}}^m f(s) (m - s) ds}{\int_{m-\underline{m}}^m f(s) ds} \right] dm + h_w(m^*) \left[\frac{\int_{m^*-\underline{m}}^{m^*} f(s) (m^* - s) ds}{\int_{m^*-\underline{m}}^{m^*} f(s) ds} \right], \quad (14)$$

which implies an average withdrawal size $W = m^* - \underline{M}$.

Computation of γ_n and $\tilde{\gamma}_n$. The statistics γ_n and $\tilde{\gamma}_n$, which measure the share of purchases settled with cards (both overall and when both payment methods were available) can be

computed through a slight modification of (10) and (11). The only difference is that now we don't consider the size of each purchase when computing the share. The card share of purchases γ is given by

$$\gamma_n = \frac{\lambda \phi \left(\int_{\underline{m}}^{m^*} h(m) \gamma_n(m) dm + h(m^*) \gamma_n(m^*) \right)}{e}, \quad (15)$$

where $\gamma_n(m) = \int_0^m f(s) p(m, s) ds + (1 - F(m))$ is the share of purchases paid with cards when having m units of cash on hand. We also want to capture how often cards are used conditional on having both options available, i.e., for *unforced* purchases. The card share of unforced purchases $\tilde{\gamma}_n$ is computed as

$$\tilde{\gamma}_n = \frac{\lambda \phi \left(\int_{\underline{m}}^{m^*} h(m) \left(\int_0^m f(s) p(m, s) ds \right) dm + h(m^*) \left(\int_0^{m^*} f(s) p(m^*, s) ds \right) \right)}{\lambda \phi \left(\int_{\underline{m}}^{m^*} h(m) \left(\int_0^m f(s) ds \right) dm + h(m^*) \left(\int_0^{m^*} f(s) ds \right) \right)}. \quad (16)$$

B.8 Card usage probabilities as a function of m , s and $m' = m - s$

We can compute the probability of cards being used as a function of some important determinants of payment choices, namely m , s and $m' = m - s$. Computing such objects requires knowing i) the invariant distribution of cash holdings $h(m)$, ii) the size distribution of payments $f(s)$, and iii) payment choice probabilities $p(m, s)$. For each of these, we can compute both the *overall* probability of card usage and the probability *conditional* on having both options available (for *unforced* purchases). We start from $\text{Pr}_{\text{ccard}}(s)$, the probability of a card payment when agents face a purchase of size s , which is given by

$$\text{Pr}_{\text{ccard}}(s) = \phi \left(H(s) + \int_s^{m^*} h(m) p(m, s) dm + h(m^*) p(m^*, s) \right), \quad (17)$$

where H is the cdf associated with the invariant distribution of cash holdings h . Cards are used to settle a purchase of size s whenever i) they are accepted, and ii) either cash holdings are not enough to cover for the transaction, or they are sufficient but cards are chosen nonetheless. The probability of card usage to settle a purchase of size s for an unforced purchase is instead

given by

$$\widetilde{\text{Pr}}_{\text{card}}(s) = \frac{\phi \left(\int_s^{m^*} h(m)p(m, s)dm + h(m^*)p(m^*, s) \right)}{\phi(1 - H(s))}. \quad (18)$$

Similarly, the unconditional probability $\text{Pr}_{\text{card}}(m)$ of using cards when having m cash balances on hand is given by

$$\text{Pr}_{\text{card}}(m) = \phi \left(\int_0^m f(s)p(m, s)ds + (1 - F(m)) \right), \quad (19)$$

while its counterpart for unforced purchases is given by

$$\widetilde{\text{Pr}}_{\text{card}}(m) = \frac{\phi \int_0^m f(s)p(m, s)ds}{\phi F(m)}. \quad (20)$$

Finally, we can write $\text{Pr}_{\text{card}}(m')$, the probability of card usage conditional on implied cash remaining in case of a cash payment being equal to $m' = m - s > 0$. Notice that since we focus on $m' > 0$, we only compute this statistic for unforced purchases, i.e.,

$$\widetilde{\text{Pr}}_{\text{card}}(m') = \frac{\phi \left(\int_{m'}^{m^*} h(m)f(m - m')p(m, m - m')dm + h(m^*)f(m^* - m')p(m^*, m^* - m') \right)}{\phi \left(\int_{m'}^{m^*} h(m)f(m - m')dm + h(m^*)f(m^* - m') \right)}. \quad (21)$$

C Calibration details

C.1 Withdrawal frequency

As shown in [Table 1](#), the probability that respondents report to have withdrawn cash during the diary day rises strongly over the four survey waves. As a result, the estimated average number of withdrawals per year increases from about 40 in 2015-16 to around 95 in 2023-24 (see [Table 4](#)). This large increase could result from changes in the questionnaire methodology over time. In SUCH and SPACE 2019, consumers were asked how much cash they added to their wallets during the diary day; we use those responses to generate a dummy for whether a withdrawal took place or not. Since SPACE 2021-22, respondents are explicitly asked “*Did you do any of the following on [DIARY DAY]? (1) Cash withdrawal (e.g. ATM machines, bank*

TABLE 4: All statistics, by wave.

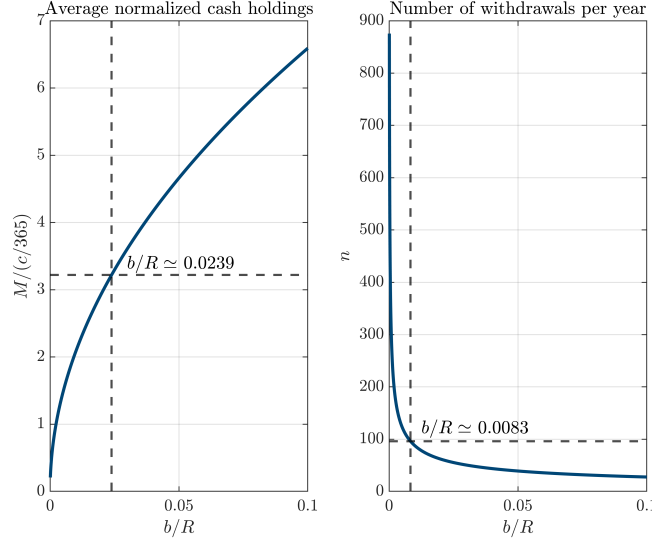
	(1)	(2)	(3)	(4)
	2015-16	2019	2021-22	2023-24
<i>Expenditure stream</i>				
Number of payments per day $\hat{\lambda}$	1.83	1.69	1.93	1.90
Daily expenditure $e/365$ (EUR)	34.03	46.28	63.89	61.23
Daily cash expenditure $c/365$ (EUR)	18.31	21.96	27.15	25.48
<i>Cash holdings</i>				
Cash holdings M (EUR)	59.85	82.55	84.98	81.93
Normalized cash holdings $M/(e/365)$	1.76	1.78	1.33	1.34
Normalized cash holdings $M/(e/365)$ (median)	0.95	0.86	0.78	0.95
Normalized cash holdings $M/(c/365)$	3.27	3.77	3.17	3.22
<i>Withdrawals</i>				
Number of withdrawals per year n	39.75	46.26	80.81	96.63
Implied number of withdrawals per year n	95.92	88.90	91.08	96.40
Withdrawal size W (EUR)	69.66	90.15	108.81	96.47
Normalized withdrawal size W/M	1.16	1.09	1.28	1.18
<i>Payment choice sets</i>				
Unforced, % share of transactions	0.63	0.66	0.66	0.73
Forced cash, % share of transactions	0.28	0.20	0.15	0.11
Forced card, % share of transactions	0.09	0.14	0.19	0.16
Unforced, % share of expenditure	0.54	0.51	0.52	0.62
Forced cash, % share of expenditure	0.17	0.14	0.11	0.08
Forced card, % share of expenditure	0.29	0.35	0.36	0.30
<i>Payment choices</i>				
Paid card, % share of transactions γ_n	0.25	0.32	0.44	0.49
Paid card (if card possible), % share of transactions $\dot{\gamma}_n$	0.35	0.40	0.51	0.55
Paid card (if unforced), % share of transactions $\tilde{\gamma}_n$	0.26	0.27	0.37	0.45
Paid card, % share of expenditure γ	0.46	0.53	0.58	0.58
Paid card (if card possible), % share of expenditure $\dot{\gamma}$	0.56	0.61	0.65	0.64
Paid card (if unforced), % share of expenditure $\tilde{\gamma}$	0.32	0.35	0.41	0.46

counter, supermarket or shop)". To check the accuracy of the withdrawal frequency measure and ensure comparability across survey waves, we adopt the following procedure. We rely on the accounting identity

$$nW = (1 - \gamma)e, \quad (22)$$

that states that, on aggregate, the average number of withdrawals per year multiplied by the mean withdrawal size should be equal to the yearly cash expenditure. We assume that

FIGURE C.1: The simple BT model and SPACE data for 2023-24.



Note: The figure shows, for a wide range of values of the ratio b/R , the values of model-implied moments M_c and n in the simple BT framework. In each plot, we display the value of b/R that enables the BT model to match the empirical counterparts $\bar{M}_c$ and $\hat{n}$ (calculated on 2023-24 SPACE data).

$\{W, \gamma, e\}$ are measured accurately and compute

$$n_{implied} = \frac{(1 - \hat{\gamma})\hat{e}}{\hat{W}},$$

which we report in [Table 4](#) as well. For the last available wave, the number of withdrawals implied by the accounting identity is remarkably close to the actual one. For SPACE 2021-22, it is slightly higher, while it is much higher for the first two waves. Over time, the implied number of withdrawals is much more comparable. Therefore, we use the implied number of withdrawals in our calibrations, for all waves and all demographic subgroups.

C.2 A benchmark: the standard BT model

We now discuss the calibration of a simple BT model to 2023-24 data, and compare the estimated total cost of managing consumption transactions that one would derive from calibrating a simple BT model with our estimates based on the augmented model.

Calibrating the BT model. Consider the canonical BT model with parameters (b, R, c) . Let $c = (1 - \gamma)e$ be annual cash expenditure, and $e = 365$ be the annual expenditure. Assume that

TABLE 5: Withdrawal cost b and total cost $\mathcal{C}^{BT}$, for different R .

Opportunity cost R	Calibration target			
	$\widehat{M}_c$	$\widehat{n}$		
	b (EUR)	$\mathcal{C}^{BT}$ (EUR)	b (EUR)	$\mathcal{C}^{BT}$ (EUR)
1%	0.015	1.677	0.005	0.990
2%	0.030	3.354	0.010	1.980
5%	0.074	8.385	0.026	4.950
10%	0.148	16.770	0.052	9.901

γ is given, and that agents finance an infinitesimal consumption stream c as in the BT model. Average cash holdings over daily cash expenditure and the yearly number of withdrawals are given by

$$M_c = \frac{M}{c/365} = 365 \frac{\sqrt{bc/2R}}{c} = 365 \sqrt{\frac{b}{2R(1-\gamma)365}} = \sqrt{\frac{365b}{2R(1-\gamma)}}, \quad (23)$$

$$n = \sqrt{\frac{Rc}{2b}} = \sqrt{\frac{R(1-\gamma)365}{2b}}, \quad (24)$$

with their ratio being given by

$$\frac{M_c}{n} = \frac{b}{R(1-\gamma)}$$

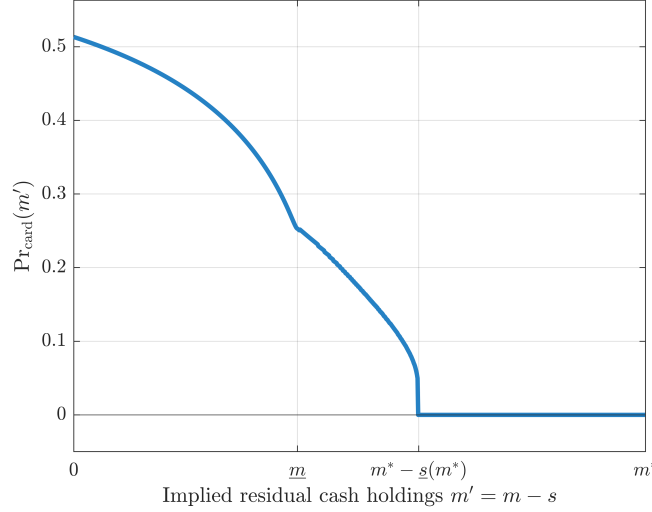
In our data for 2023-24 (the most recent wave) we have $\widehat{M}_c = 3.22$, $\widehat{\gamma} = 0.58$ and $\widehat{n} = 96.4$. Assume we target $\widehat{M}_c$; from Equation (23), this implies $\frac{b}{R} \simeq 0.0234$. Assume that instead we target $\widehat{n}$; from Equation (24) this yields $\frac{b}{R} \simeq 0.0083$. The BT model cannot fit both moments at the same time while simultaneously matching $\widehat{\gamma}$. Figure C.1 shows this graphically: the observed $\widehat{M}_c$ is consistent with a ratio b/R around 3 times higher than what the average number of withdrawals per year $\widehat{n}$ implies.

Total cost in the BT model. As an additional exercise, we compute the total cost implied by the BT model when targeting $\widehat{M}_c$ or $\widehat{n}$ respectively. Total costs of managing consumption transactions in the BT model are given by $\mathcal{C}^{BT} = RM + bn$. In Table 5, we display, for several values of R (including our benchmark of 10% used in the calibration), the value of b needed to match either cash holdings or the frequency of withdrawals, and the implied values of $\mathcal{C}^{BT}$.

When $R = 10\%$, as in our calibration in the main text, and we target average normalized cash holdings, we find a value of total costs in line with our estimates from the augmented model.

D Model and calibration: additional figures and tables

FIGURE D.1: Probability of card usage as a function of implied residual cash balances.

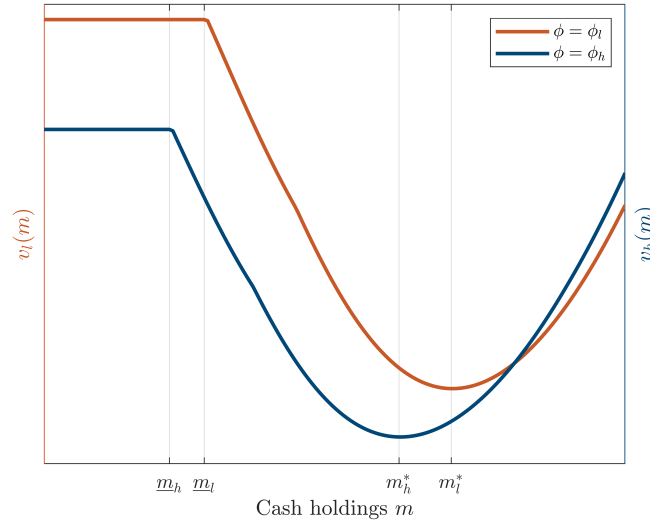


Note: The graph shows the model-implied predicted probability $\widetilde{\Pr}_{\text{card}}(m')$ of card usage as a function of implied remaining cash holdings if the purchase is settled in cash $m' = m - s$. $m^* - \underline{s}(m^*)$ is the smallest level of implied remaining cash holdings that triggers some card usage. Parameters are those obtained from the calibration of the model for 2021-22 discussed in [Section 4](#).

TABLE 6: Parameter estimates, all waves.

	Time period			
	2015-16	2019	2021-22	2023-24
<i>Externally calibrated parameters</i>				
Purchase size distr. F , location μ_s	-1.840	-1.823	-1.568	-1.955
Purchase size distr. F , scale σ_s^2	2.473	2.602	1.817	2.630
Card acceptance rate ϕ	0.722	0.802	0.848	0.894
<i>Internally calibrated parameters (minimum distance)</i>				
Withdrawal cost b (EUR)	0.005	0.006	0.013	0.003
Card usage cost κ (EUR)	0.003	0.004	0.007	0.002
Lost purchase cost u (EUR)	0.288	0.627	0.527	0.976
Purchase opport. per day $\lambda/365$	1.876	1.722	1.982	1.923

FIGURE D.2: The effect of card acceptance in the model.



Note: The figure displays the value function $v(m)$ for two values of the card acceptance rate ϕ , with ϕ_l denoting lower card acceptance and ϕ_h denoting higher card acceptance. To produce the plots, we set $\phi_l = 0.85$ and $\phi_h = 0.89$, and we keep all the other parameters at their levels obtained from the calibration of the model for 2021-22 discussed in [Section 4](#). We plot the two value functions on separate y -axes, so their levels are not comparable.

FIGURE D.3: Total cost, all waves.

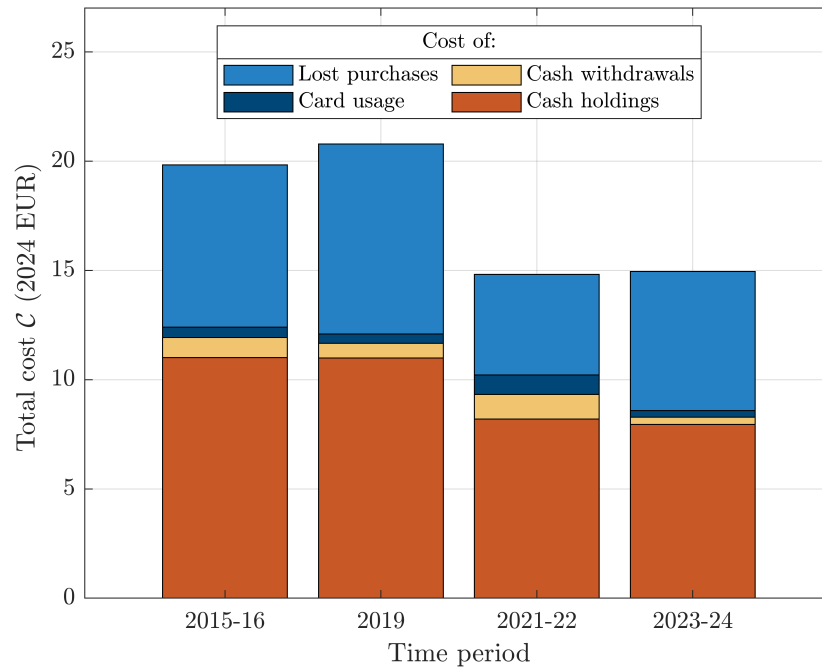


FIGURE D.4: Total cost, 2023-24, subgroups.

